



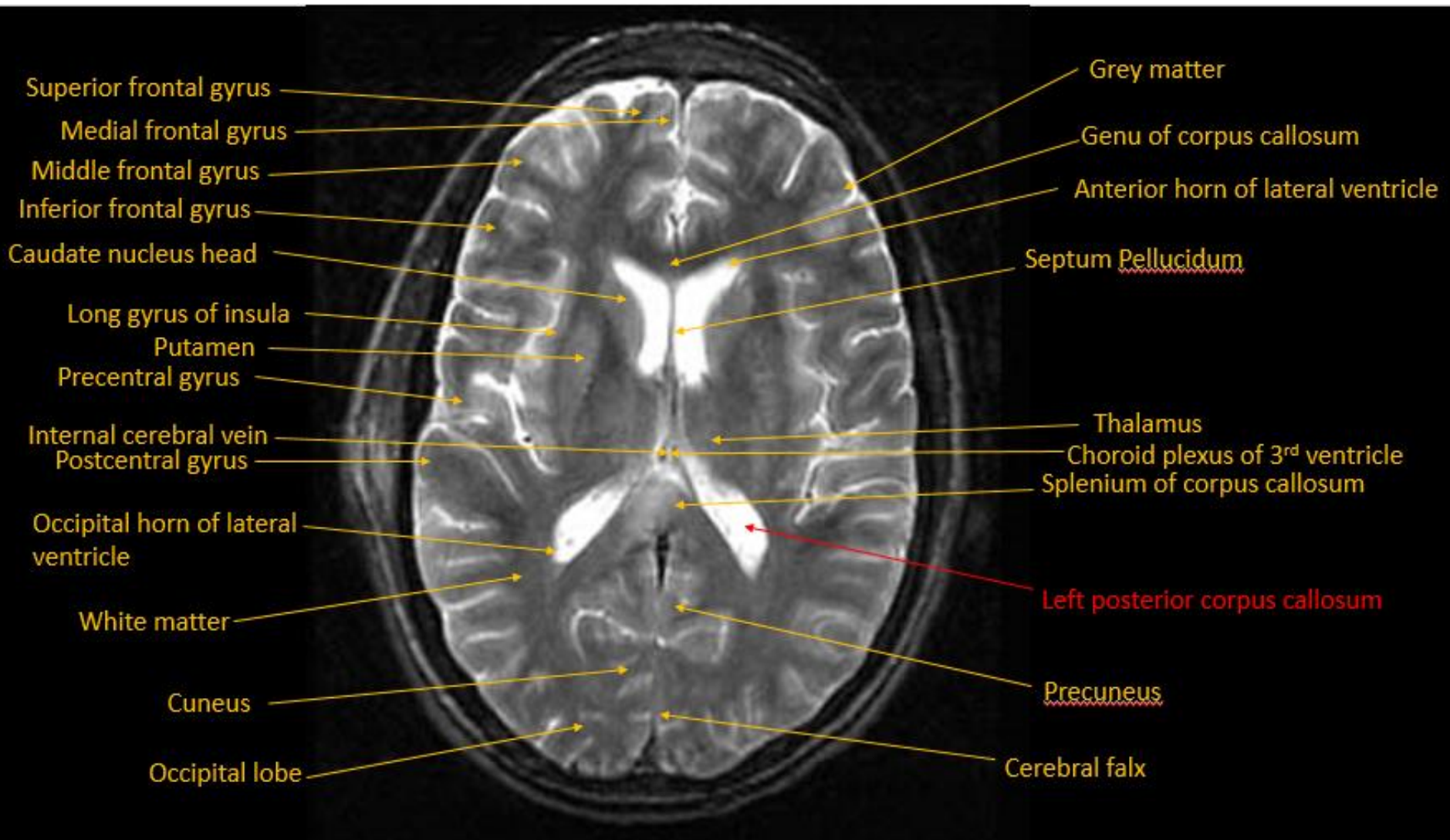
IS-NPMSD

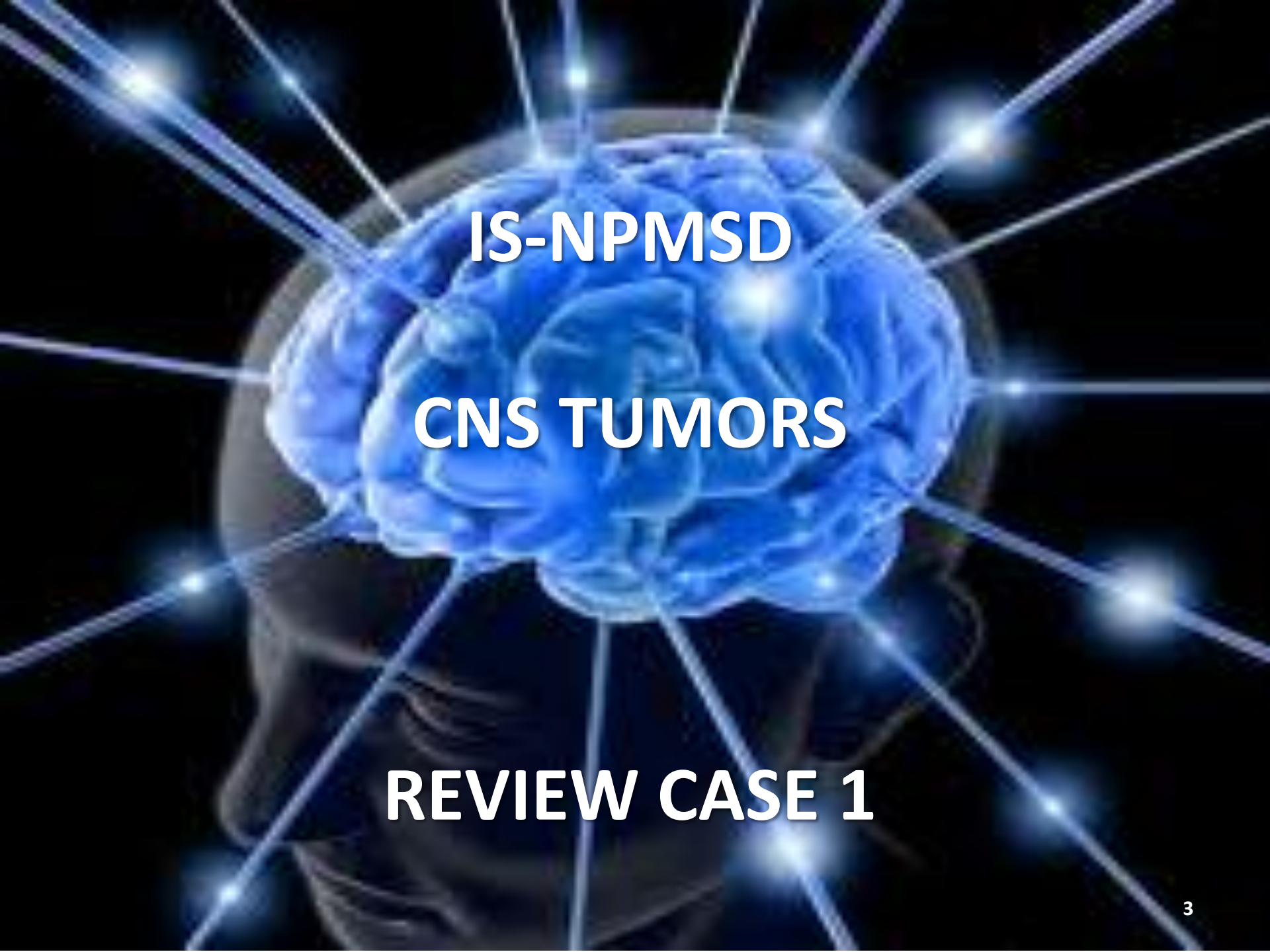
CNS TUMORS

REVIEW CASES 1-6

LIISA A. RUSSELL, MD

CNS MRI ANATOMY





IS-NPMSD

CNS TUMORS

REVIEW CASE 1

REVIEW CASE 1

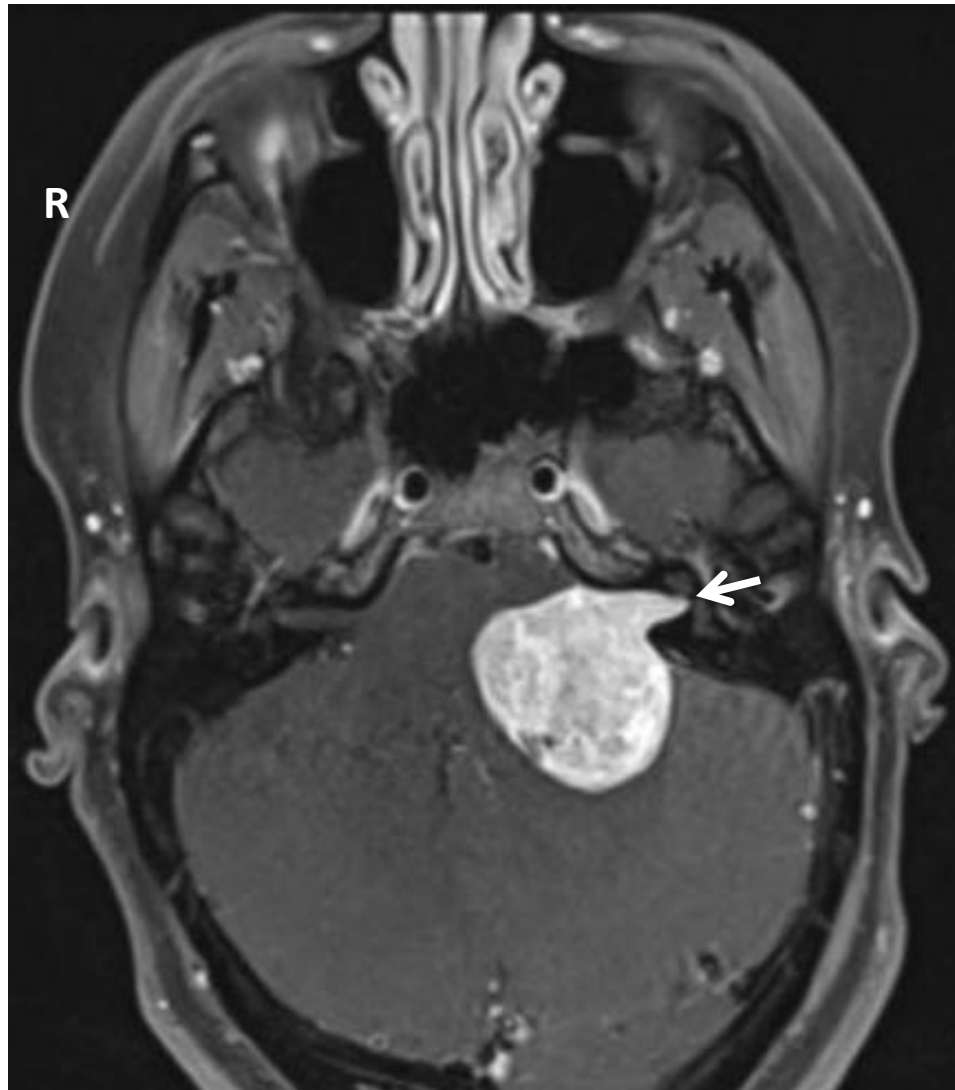
1345.svs

The patient is a 55-year-old man, who complained of vertigo and ringing in his left ear. Subsequently he lost hearing in his left ear and developed decreased function of the facial muscles and sensation on the left side of his face. A CT-scan revealed a well-circumscribed mass at the base of skull on the left side. The mass was removed.

1. Is this tumor intra- or extra-dural?
2. Is this tumor intra- or extra-axial?
3. What is your clinical differential diagnosis based on the history and imaging studies?
4. Is this tumor benign or malignant?
5. What are the diagnostic histologic features?
6. What is the diagnosis?
7. What is the significance, if a lesion with the histologic features seen in this case is bilateral?

REVIEW CASE 1

DESCRIBE THE LESION AND THE LOCATION



WHAT IS YOUR CLINICAL IMPRESSION?

REVIEW CASE 1

WHAT IS THE MOST LIKELY DIAGNOSIS?

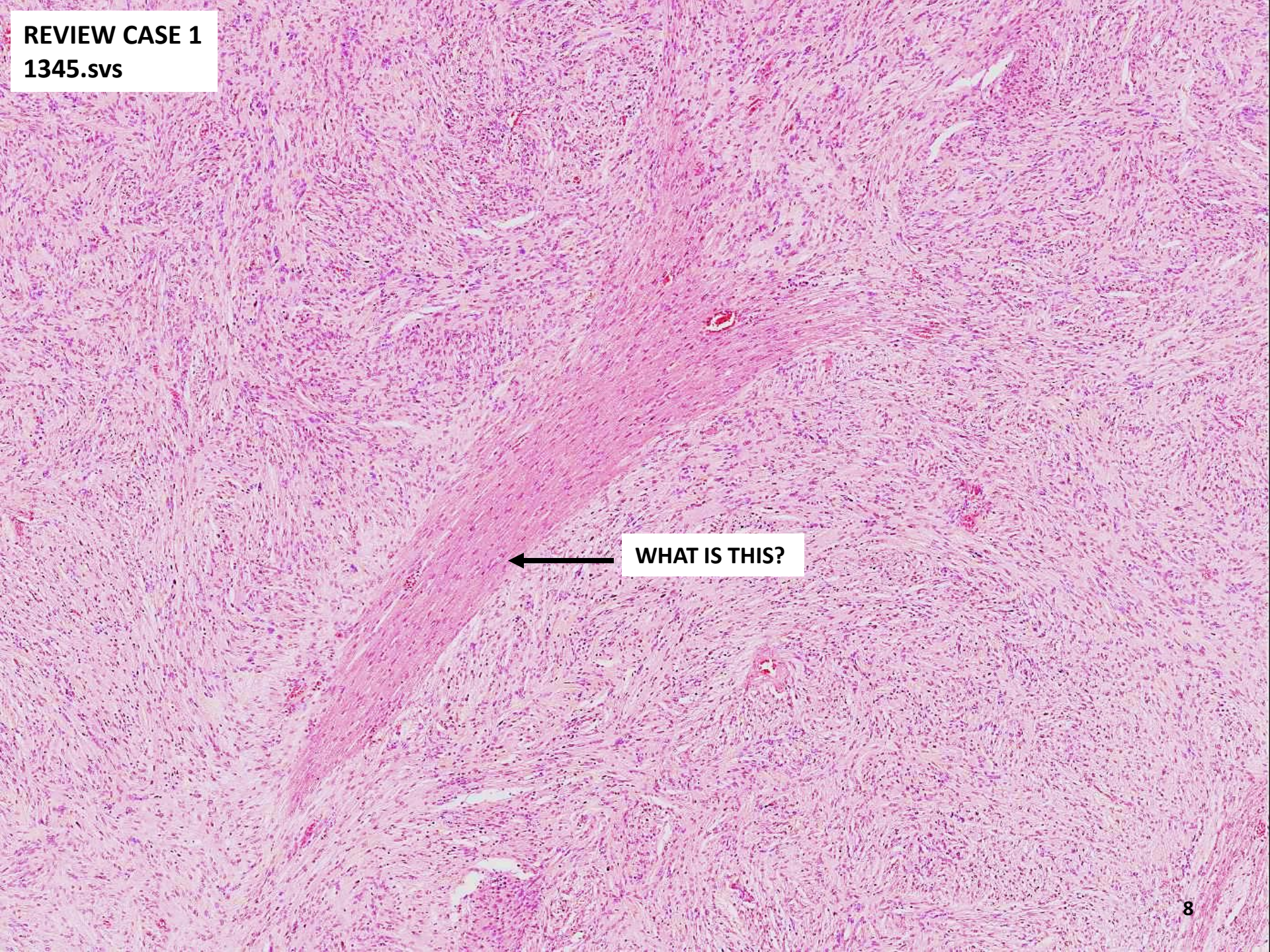
**DIFFERENT
PATIENT, SAME DISEASE**



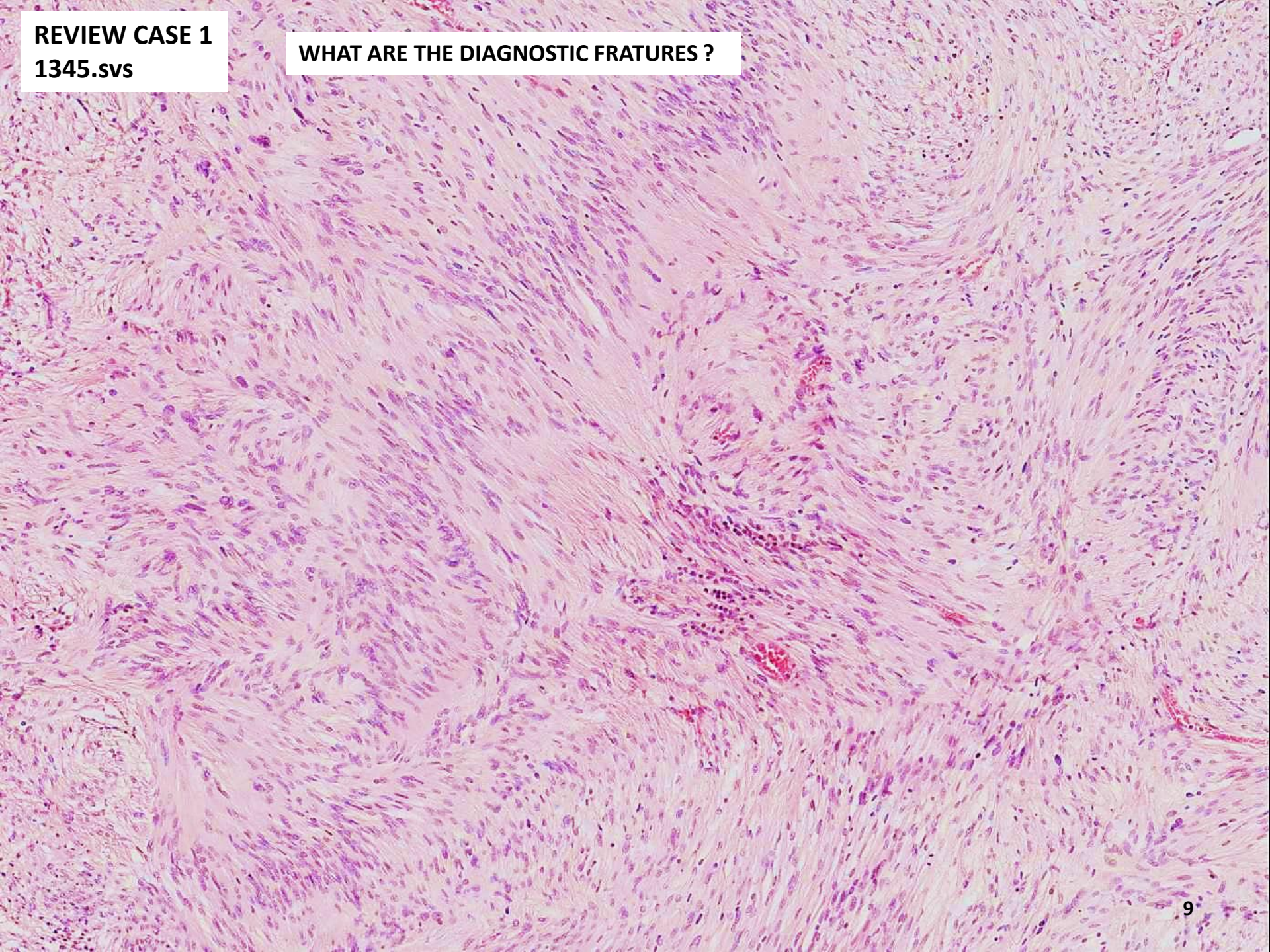
**INTRA-DURAL EXTRA-AXIAL TUMOR
IN THE CEREBELLO-PONTINE ANGLE**

WHAT IS THIS?

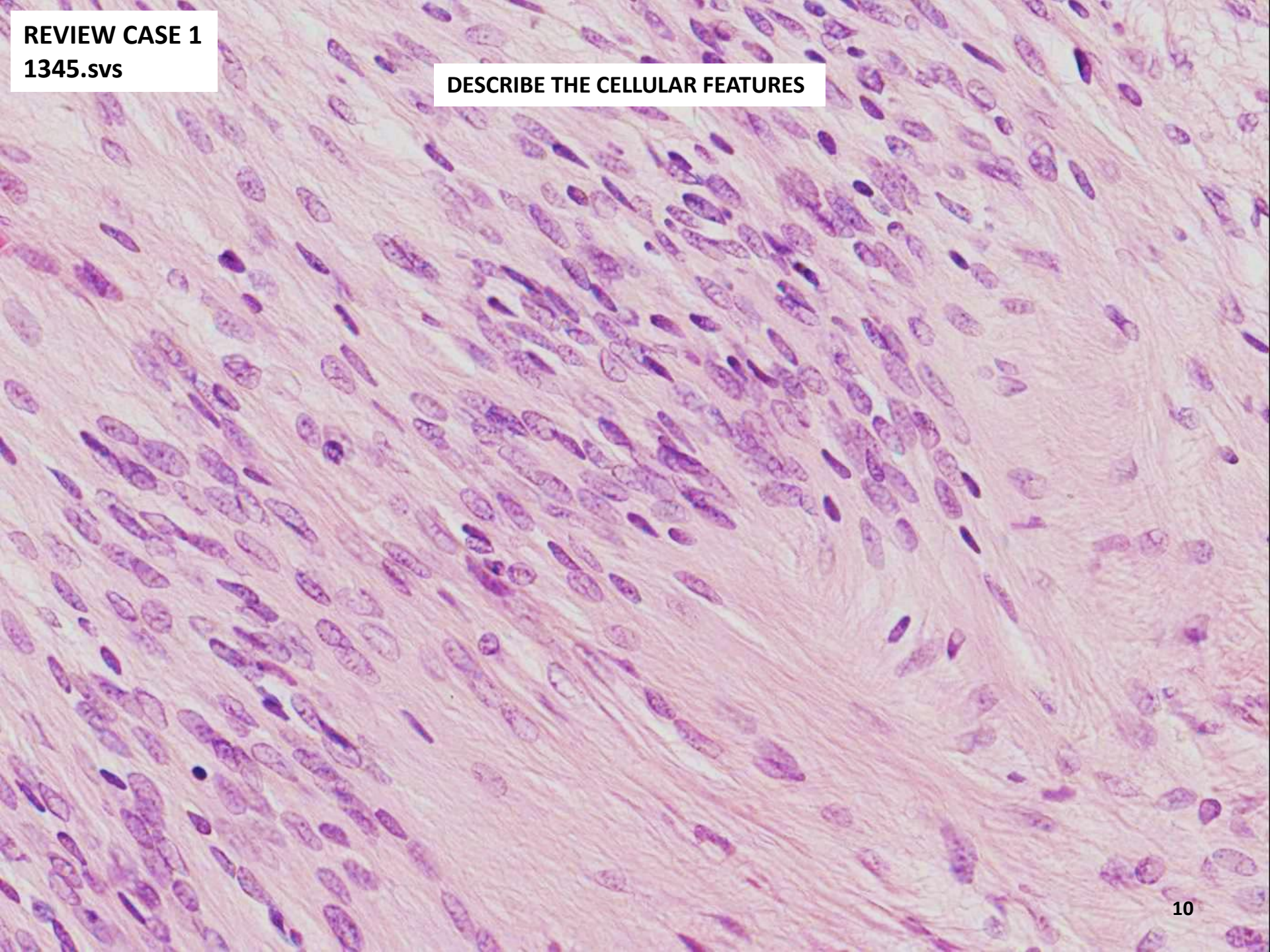


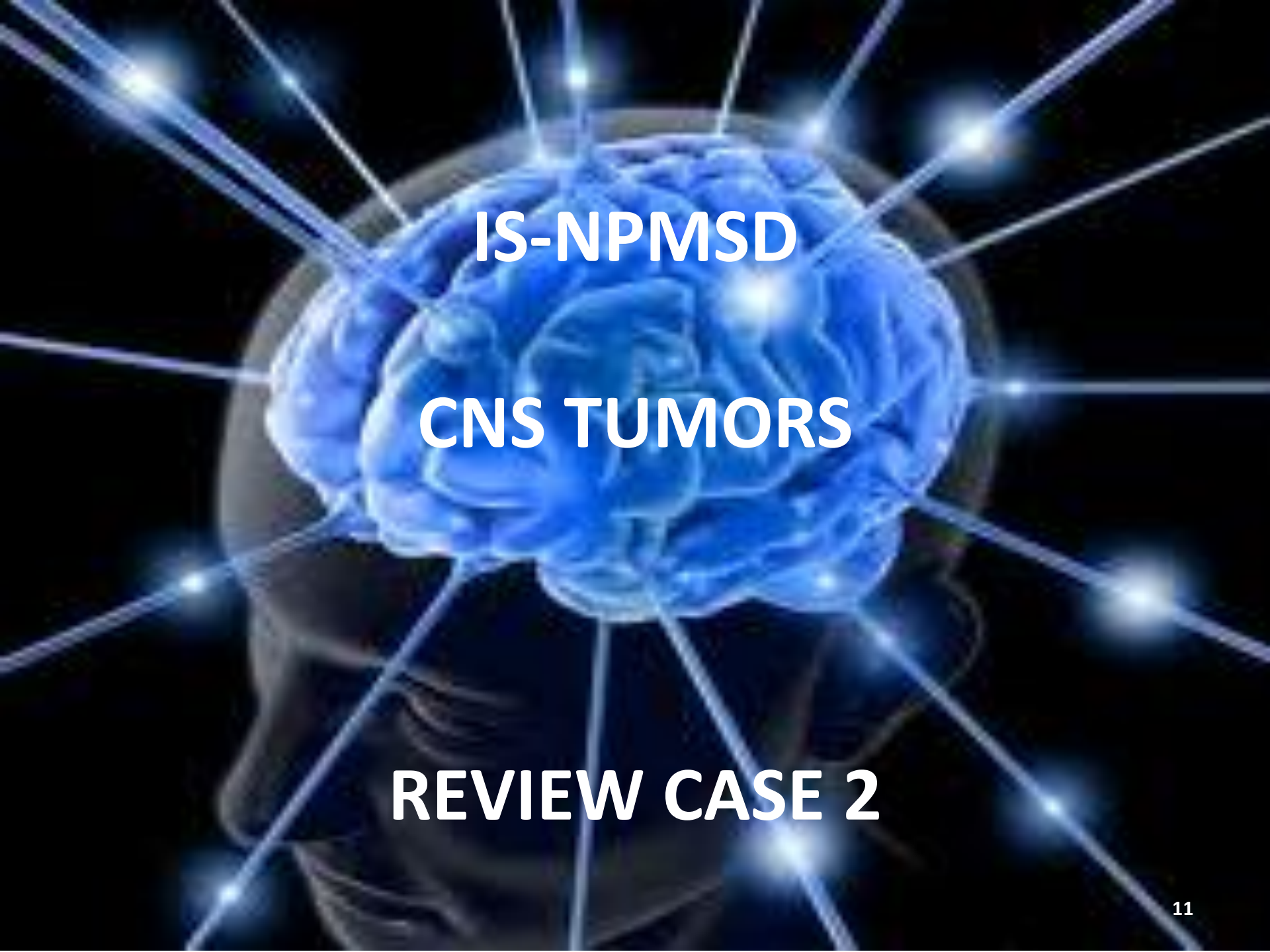


WHAT IS THIS?



DESCRIBE THE CELLULAR FEATURES





IS-NPMSD

CNS TUMORS

REVIEW CASE 2

REVIEW CASE 2

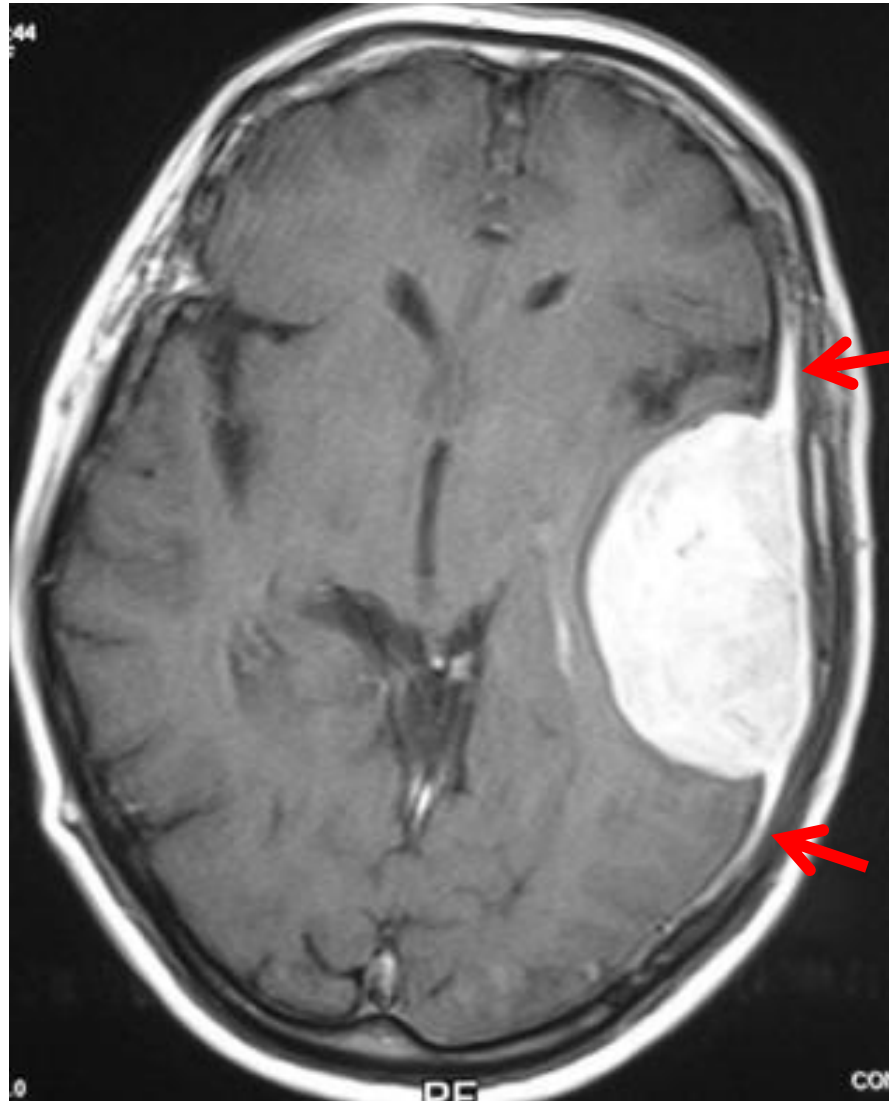
758.svs

The patient is a previously healthy 36-year-old woman, who started experiencing headaches with increasing frequency. On the morning of admission to the hospital she had a grand mal seizure and when she regained consciousness, she noted that her right leg was weak. She underwent a battery of tests and was noted to have a well-demarcated mass beneath the dura compressing the left parietal lobe. She was referred to a neurosurgeon.

- 1. Is this tumor intra- or extra-dural?**
- 2. intra- or extra-axial?**
- 3. What is your clinical differential diagnosis based on the history and imaging studies?**
- 4. Is this tumor benign or malignant?**
- 5. What are the diagnostic histologic features?**
- 6. What is the diagnosis?**
- 7. Where does the blood supply to this lesion come from?**
- 8. What is the chromosomal change typically found in lesions like this one?**
- 9. Is the patient's gender a significant factor?**

REVIEW CASE 2

T1-WEIGHTED MRI WITH CONTRAST

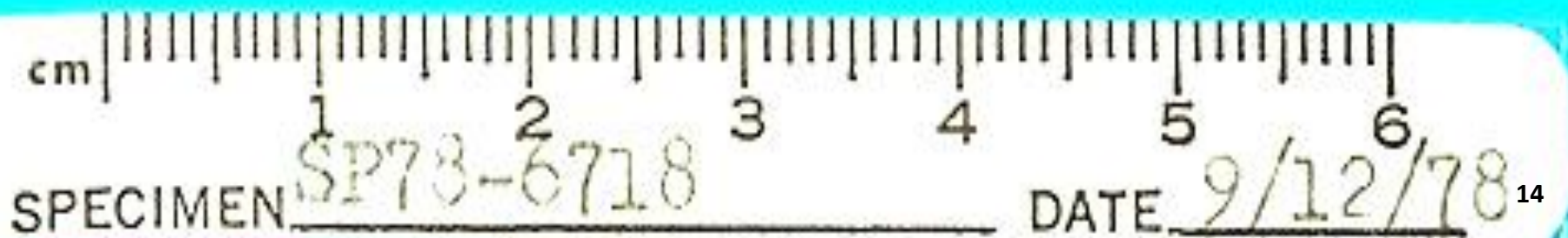
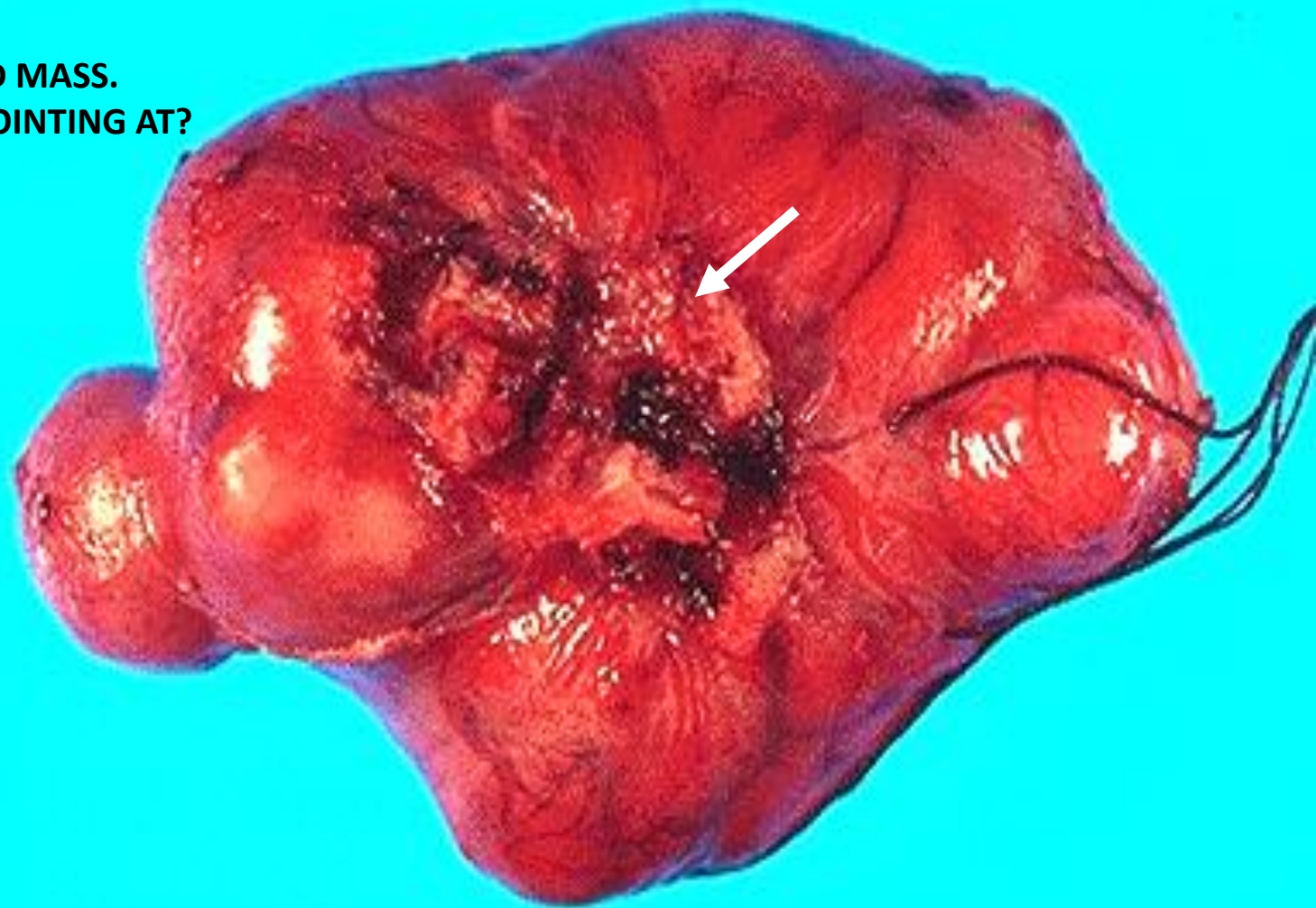


**DESCRIBE
THE FINDINGS.
WHERE IS THIS LESION?**

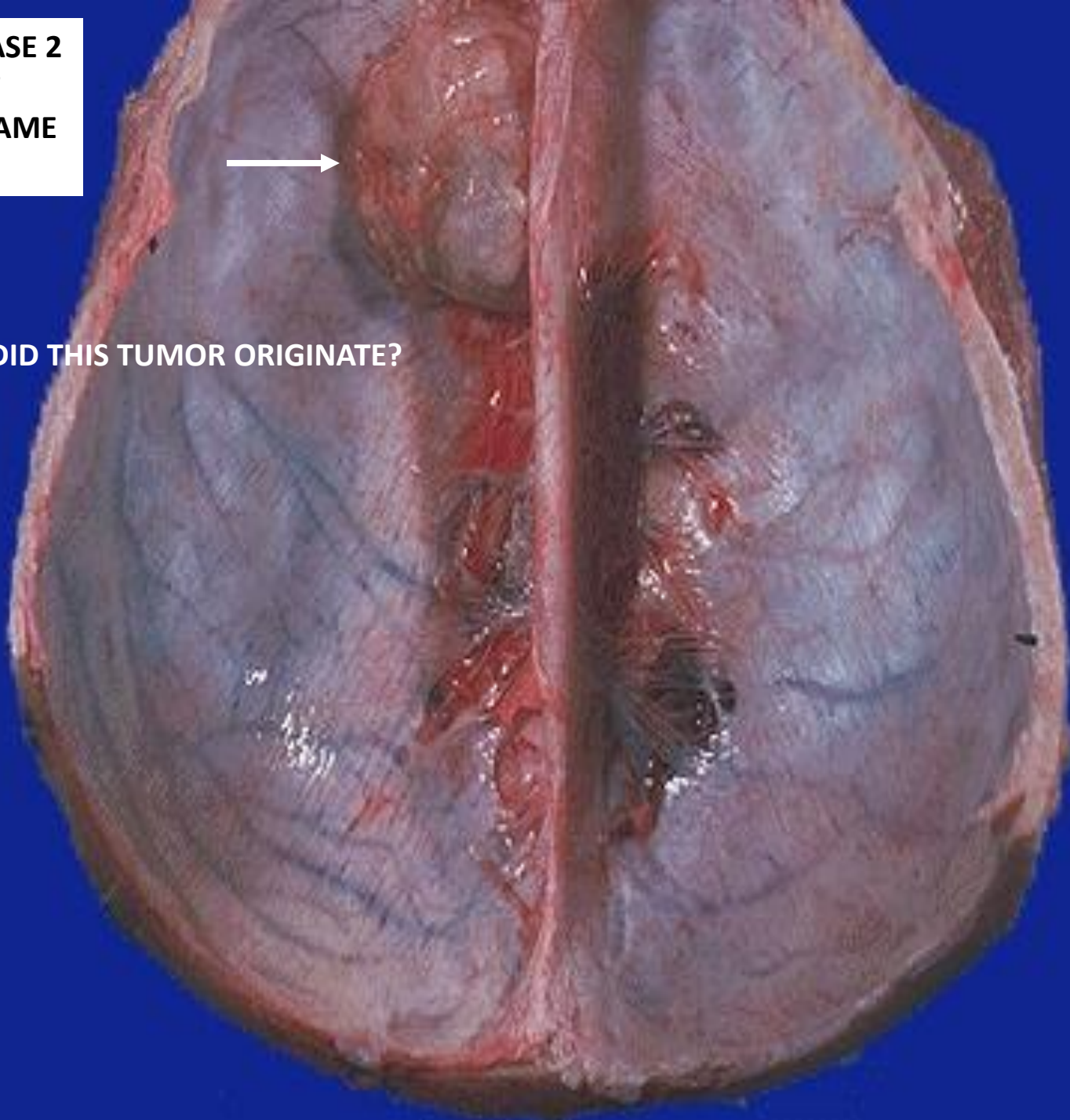
WHAT IS YOUR TENTATIVE DIAGNOSIS?

REVIEW CASE 2

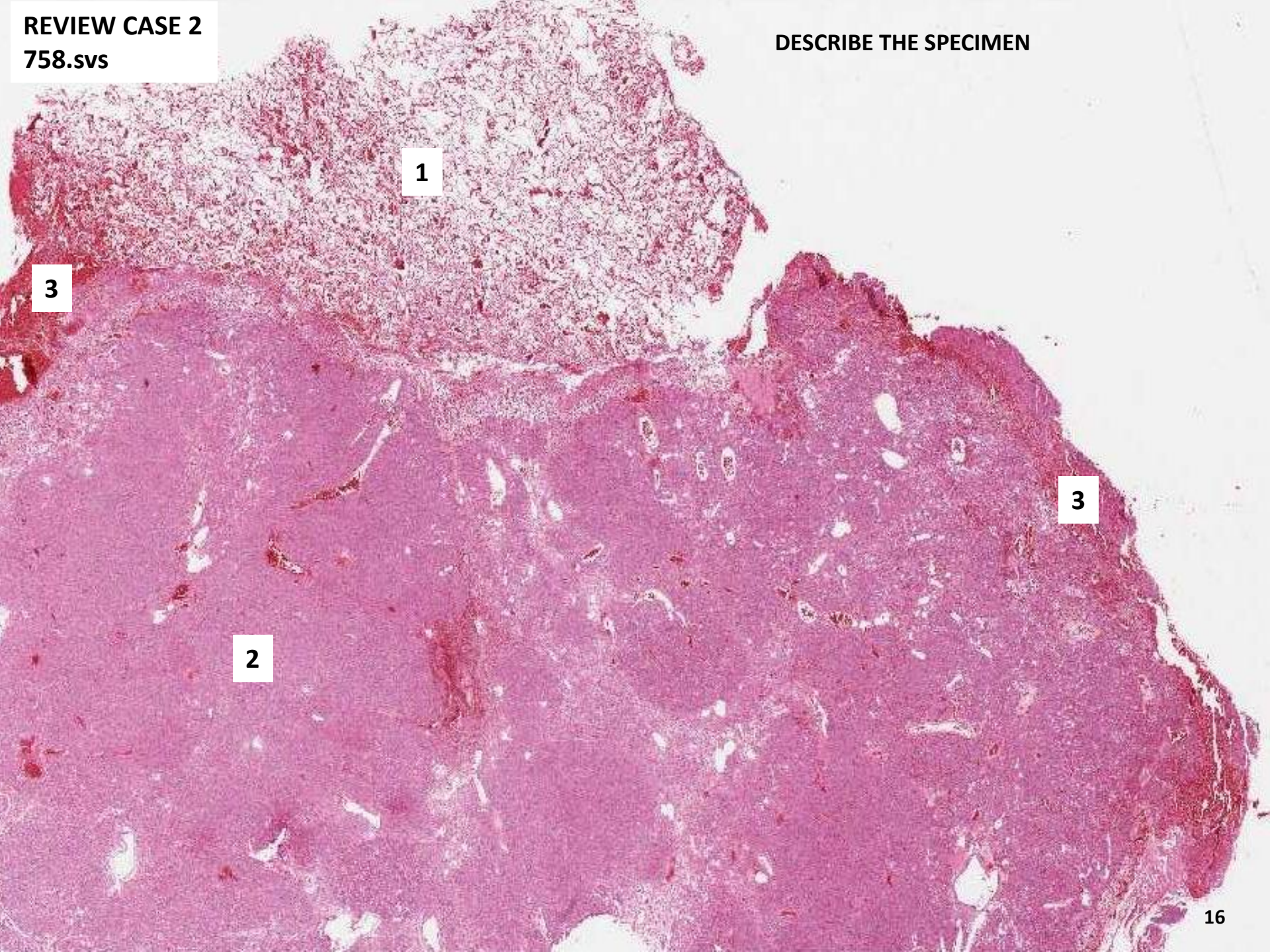
**DESCRIBE THE RESECTED MASS.
WHAT IS THE ARROW POINTING AT?**

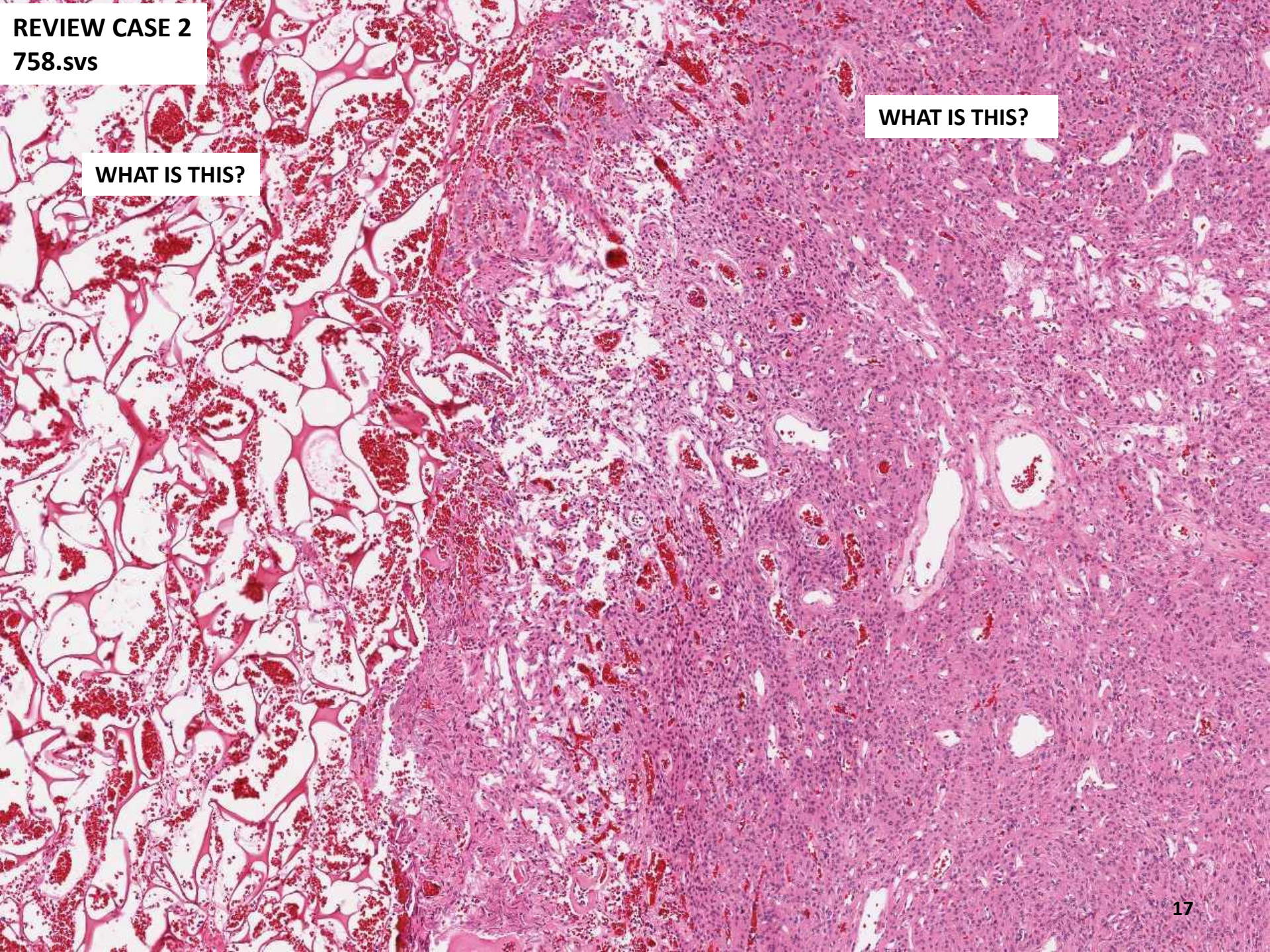


**REVIEW CASE 2
DIFFERENT
PATIENT, SAME
DISEASE**



WHERE DID THIS TUMOR ORIGINATE?

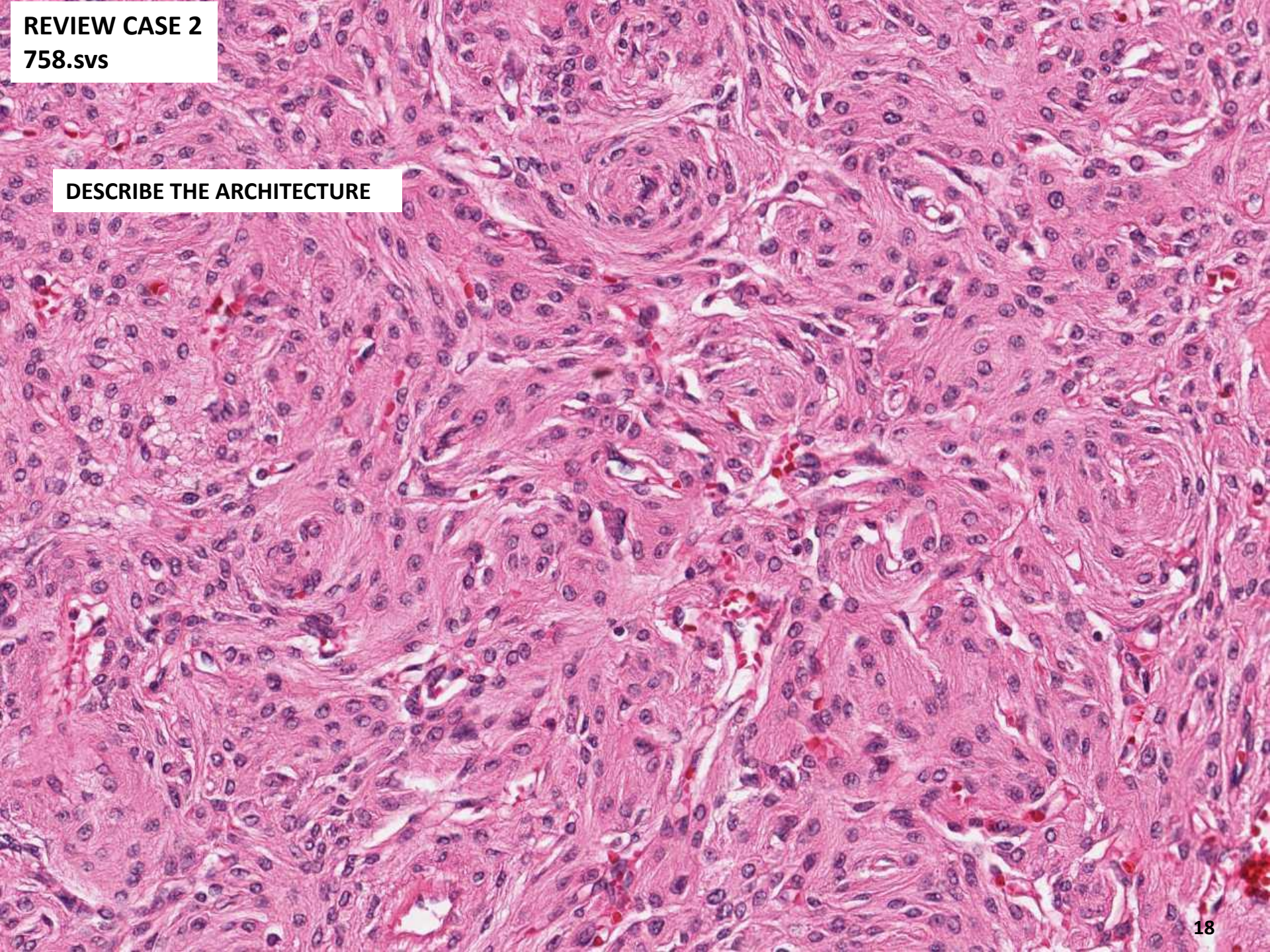




WHAT IS THIS?

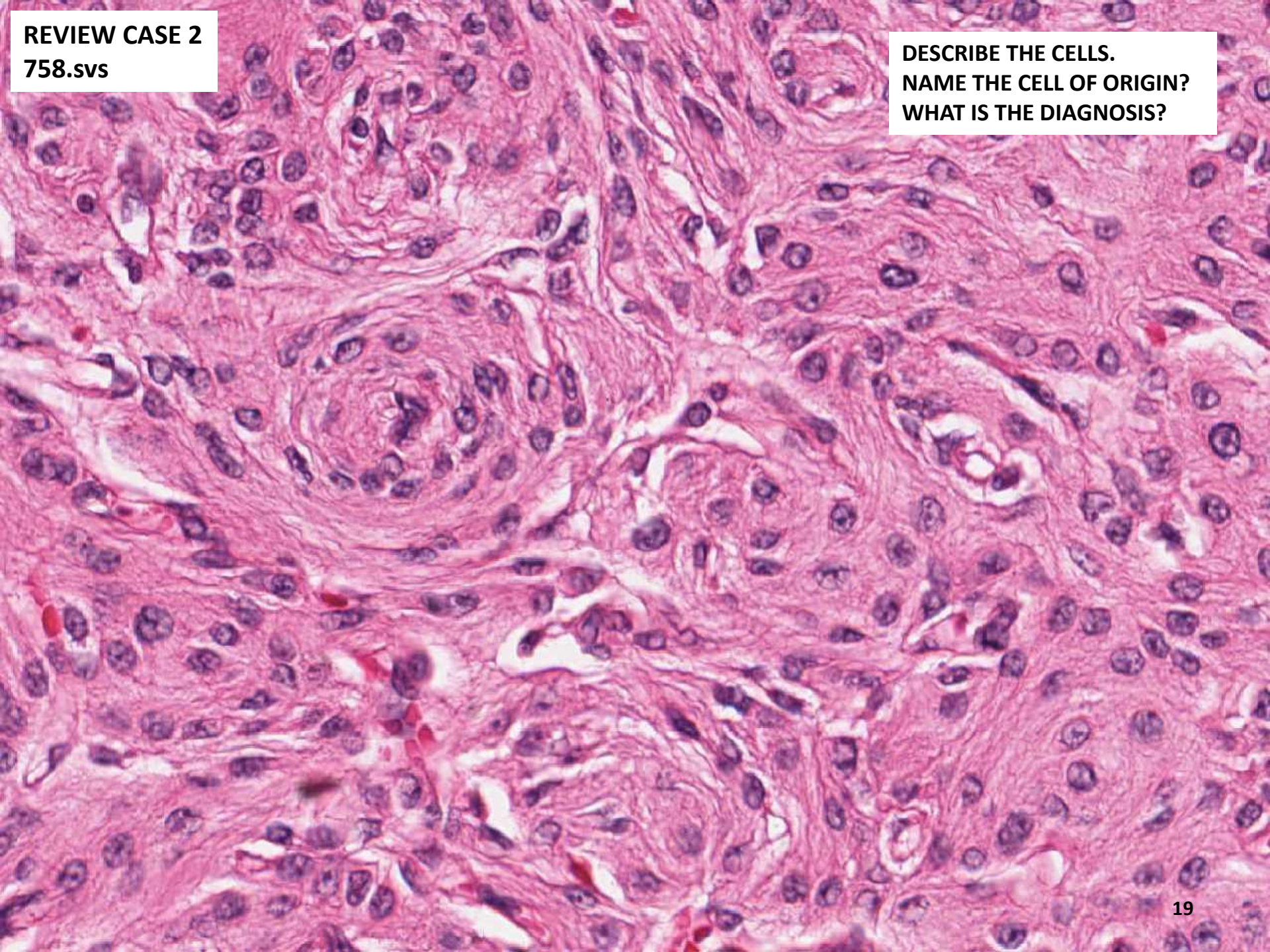
WHAT IS THIS?

DESCRIBE THE ARCHITECTURE



REVIEW CASE 2
758.svs

DESCRIBE THE CELLS.
NAME THE CELL OF ORIGIN?
WHAT IS THE DIAGNOSIS?



IS-NPMSD
CNS AND TUMORS

REVIEW CASE 3

REVIEW CASE 3

1344.svs

The patient was a 40-year-old man, who had a history of hand tremors for 8 years and then started experiencing vertigo, headache, and nausea, which was worse in the morning, and was found to have bilateral papilledema. An MRI-scan revealed a mass arising in the floor of the fourth ventricle.

- 1. Is this tumor intra- or extra-dural?**
- 2. Is this tumor intra- or extra-axial?**
- 3. What is your clinical differential diagnosis based on the history and imaging studies?**
- 4. Is this tumor benign or malignant?**
- 5. What are the diagnostic histologic features?**
- 6. What is the diagnosis?**
- 7. In which age-group is this tumor most common?**
- 8. What is the difference in the presentation between pediatric and adult cases?**

REVIEW CASE 3

NO CONTRAST

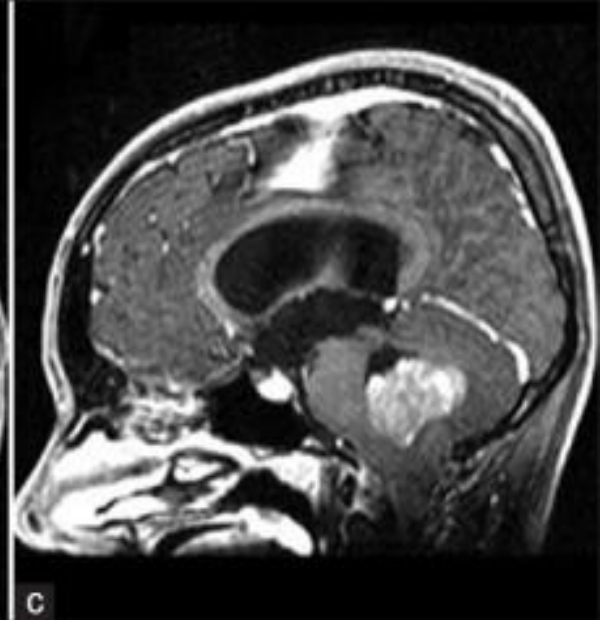
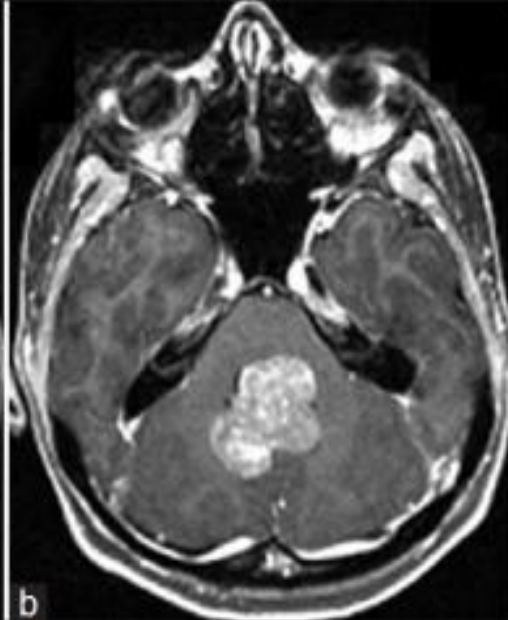
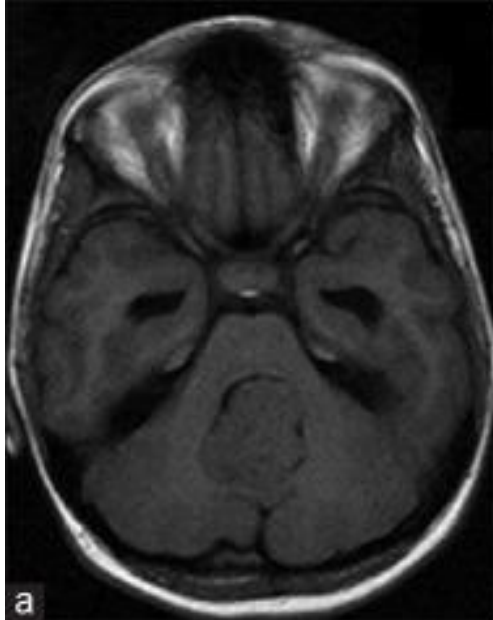
WITH CONTRAST

WHAT ARE THESE?

CT



MRI

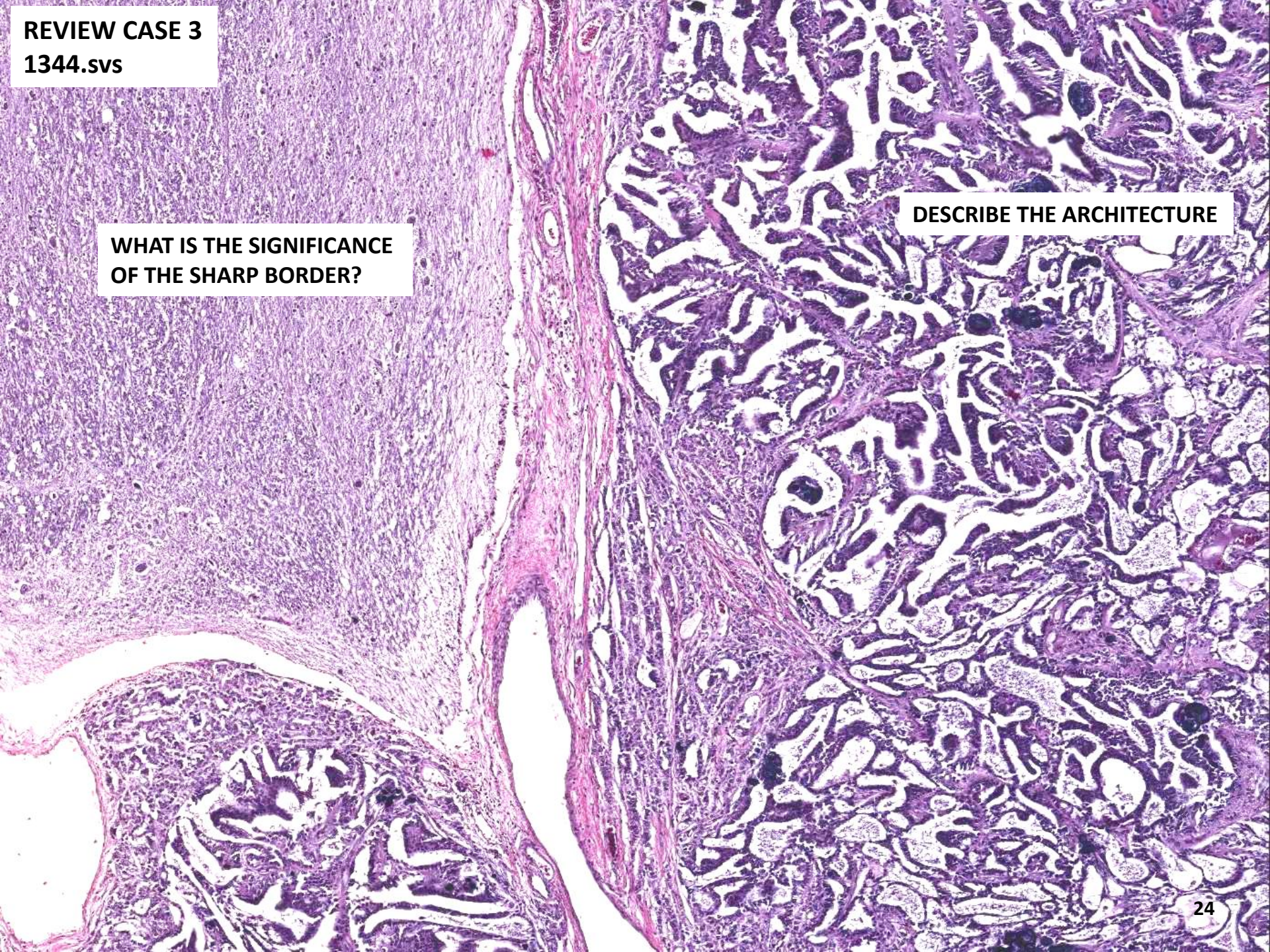


WHAT IS THIS?

WHAT IS THIS?

**WHAT IS THE SIGNIFICANCE
OF THE SHARP BORDER?**

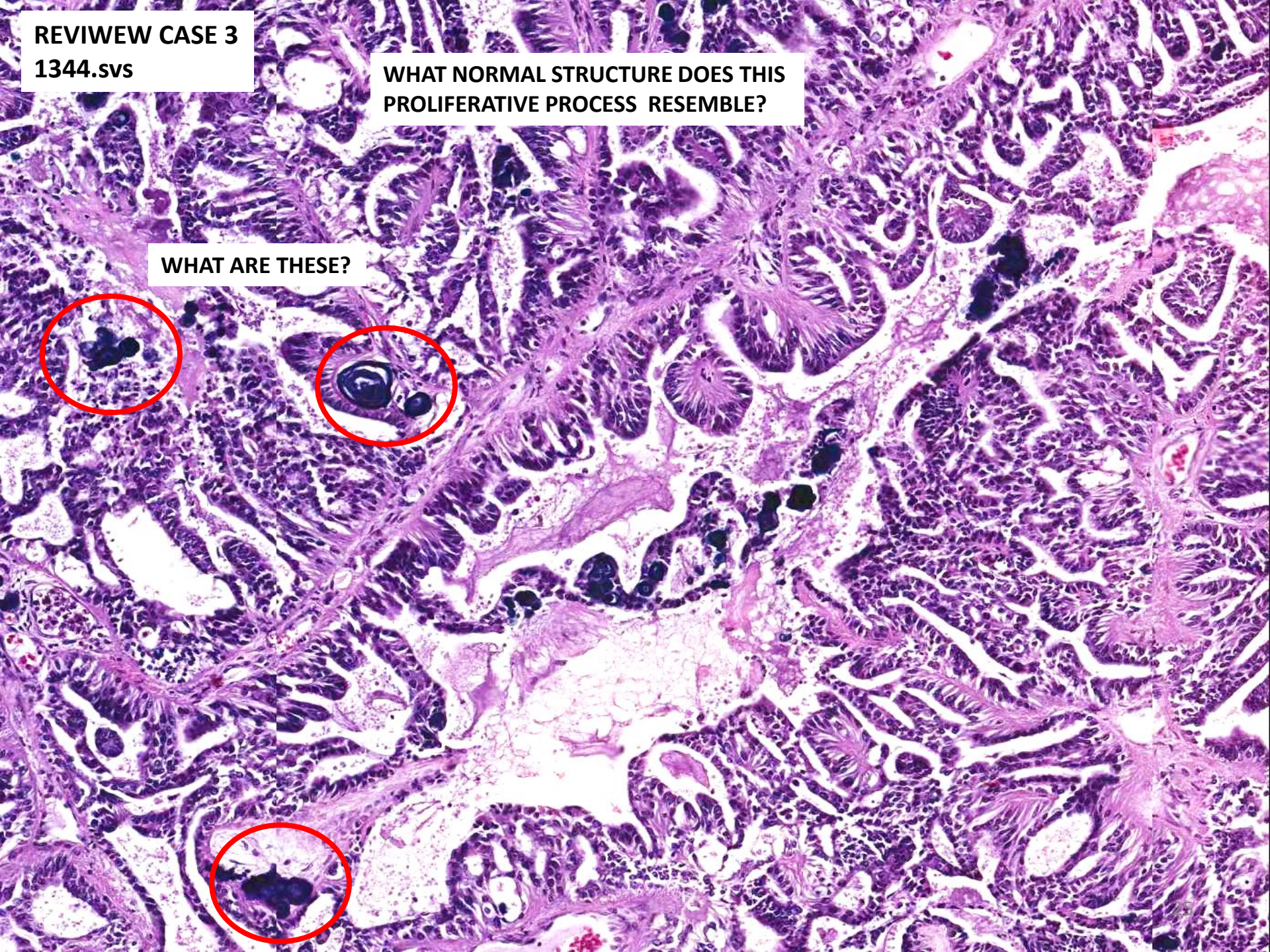
DESCRIBE THE ARCHITECTURE



REVIEW CASE 3
1344.svs

WHAT NORMAL STRUCTURE DOES THIS
PROLIFERATIVE PROCESS RESEMBLE?

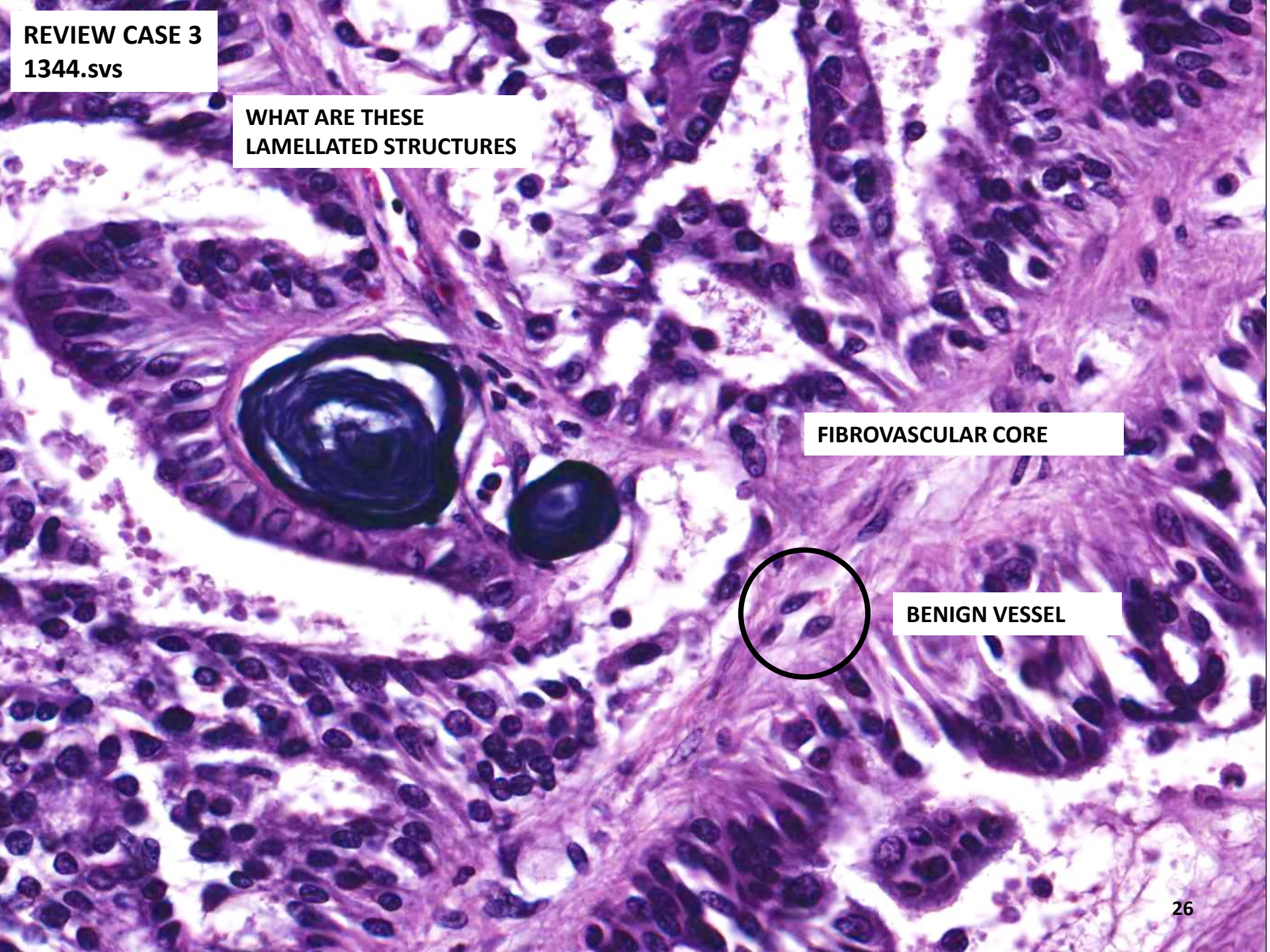
WHAT ARE THESE?

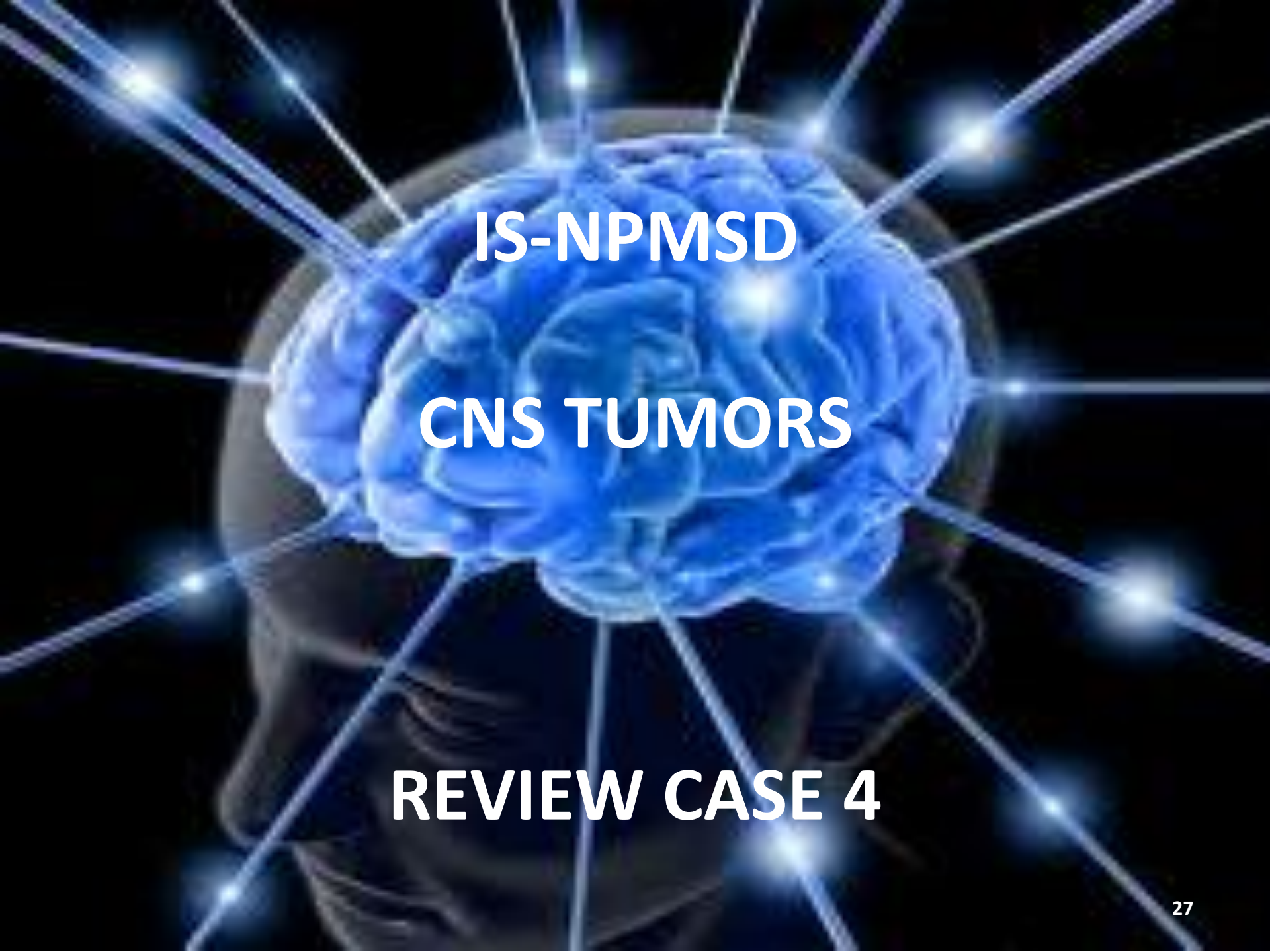


**WHAT ARE THESE
LAMELLATED STRUCTURES**

FIBROVASCULAR CORE

BENIGN VESSEL





IS-NPMSD

CNS TUMORS

REVIEW CASE 4

REVIEW CASE 4

1346.svs

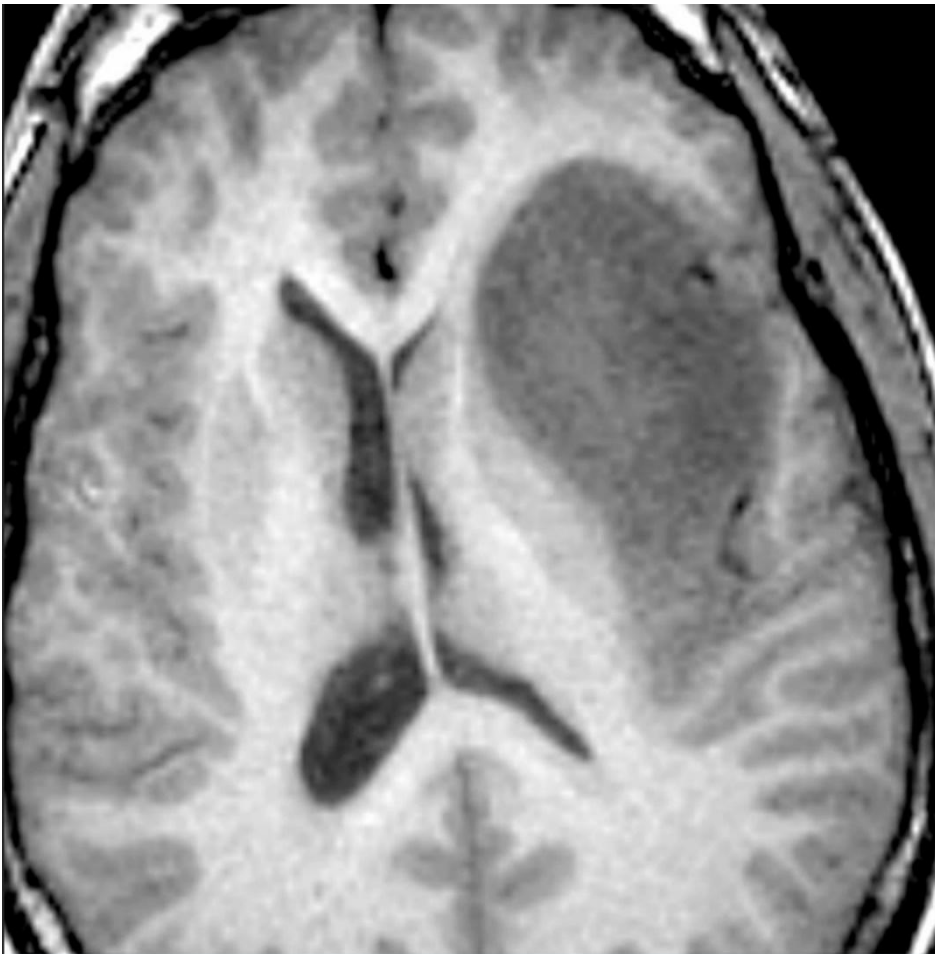
The patient was a 44-year-old woman with a history of astrocytoma, grade II, removed one and a half years previously and treated with postoperative radiation. The tumor recurred and at autopsy a large poorly demarcated gray-white mass was found in the left frontotemporal area. The horn of the lateral ventricle was completely obliterated by the tumor.

1. Is this tumor intra- or extra-dural?
2. Is this tumor intra- or extra-axial?
3. What is your clinical differential diagnosis based on the history and imaging studies?
4. Is this tumor benign or malignant?
5. What are the diagnostic histologic features?
6. What is the diagnosis/ the differential diagnosis?

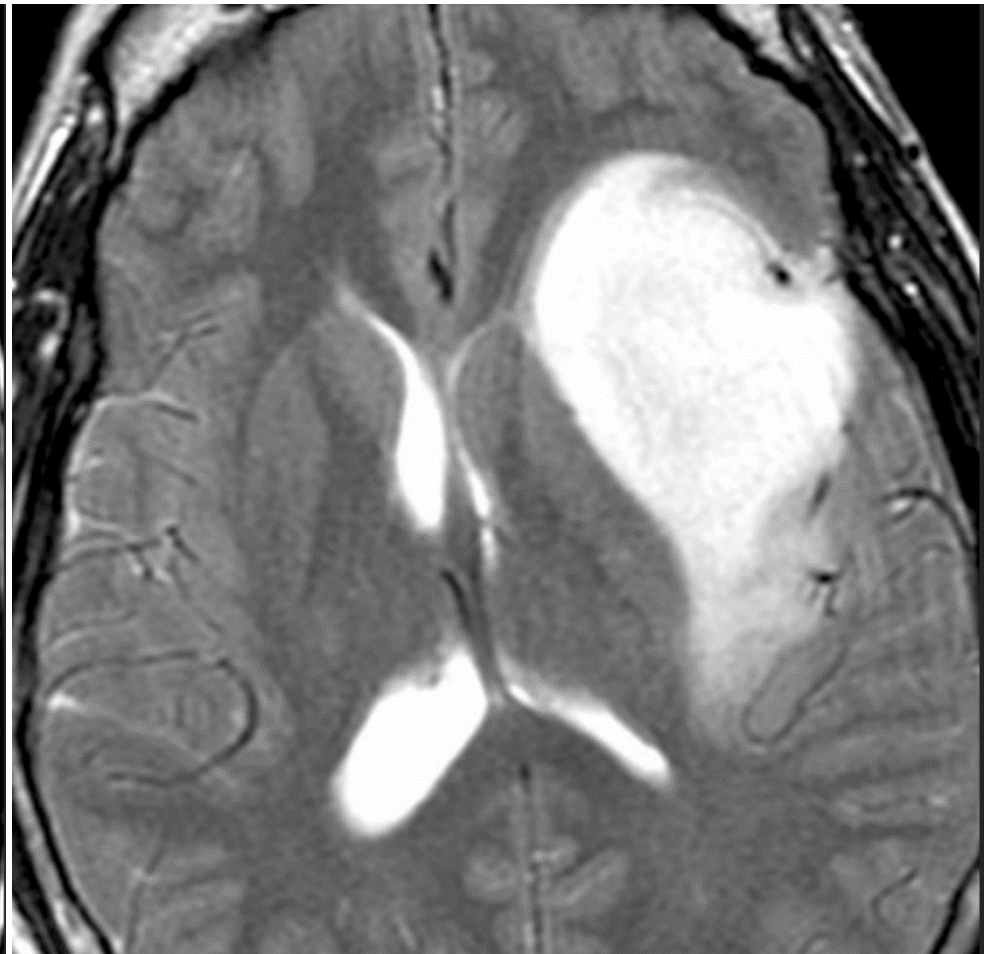
REVIEW CASE 4

MRI WITHOUT CONTRAST

T1



T2



WHAT IS THIS?

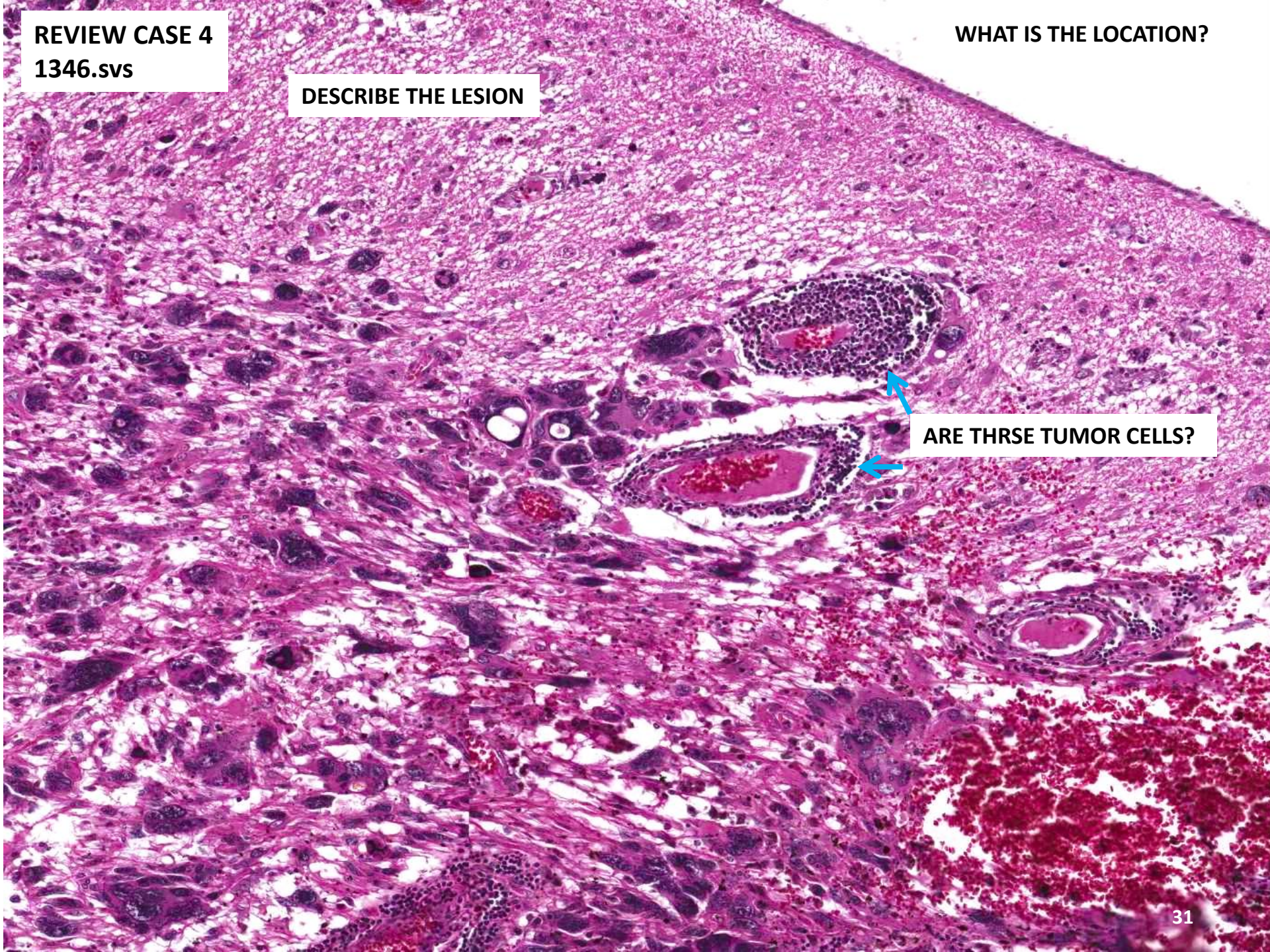


REVIEW CASE 4
1346.svs

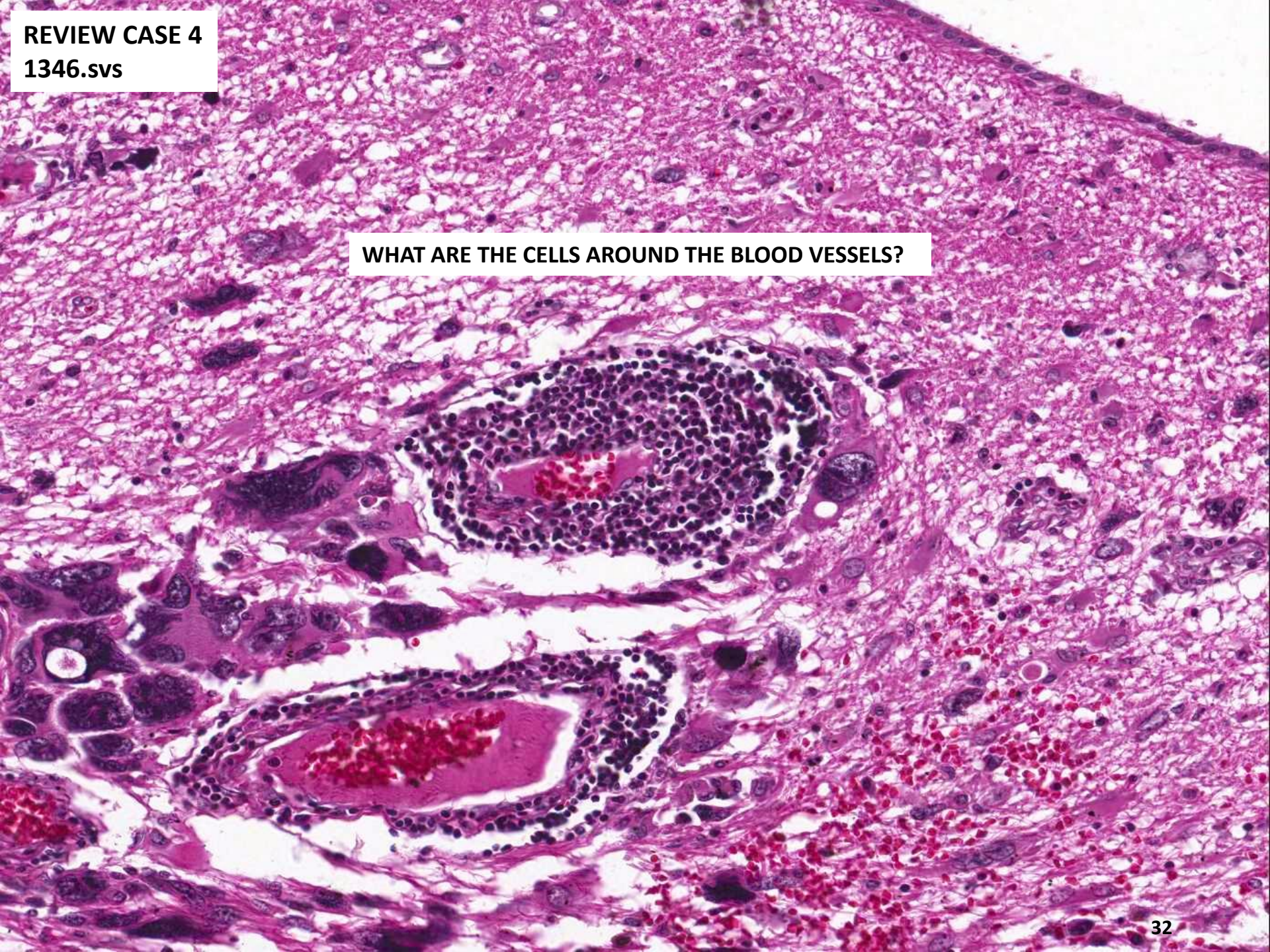
WHAT IS THE LOCATION?

DESCRIBE THE LESION

ARE THRE TUMOR CELLS?

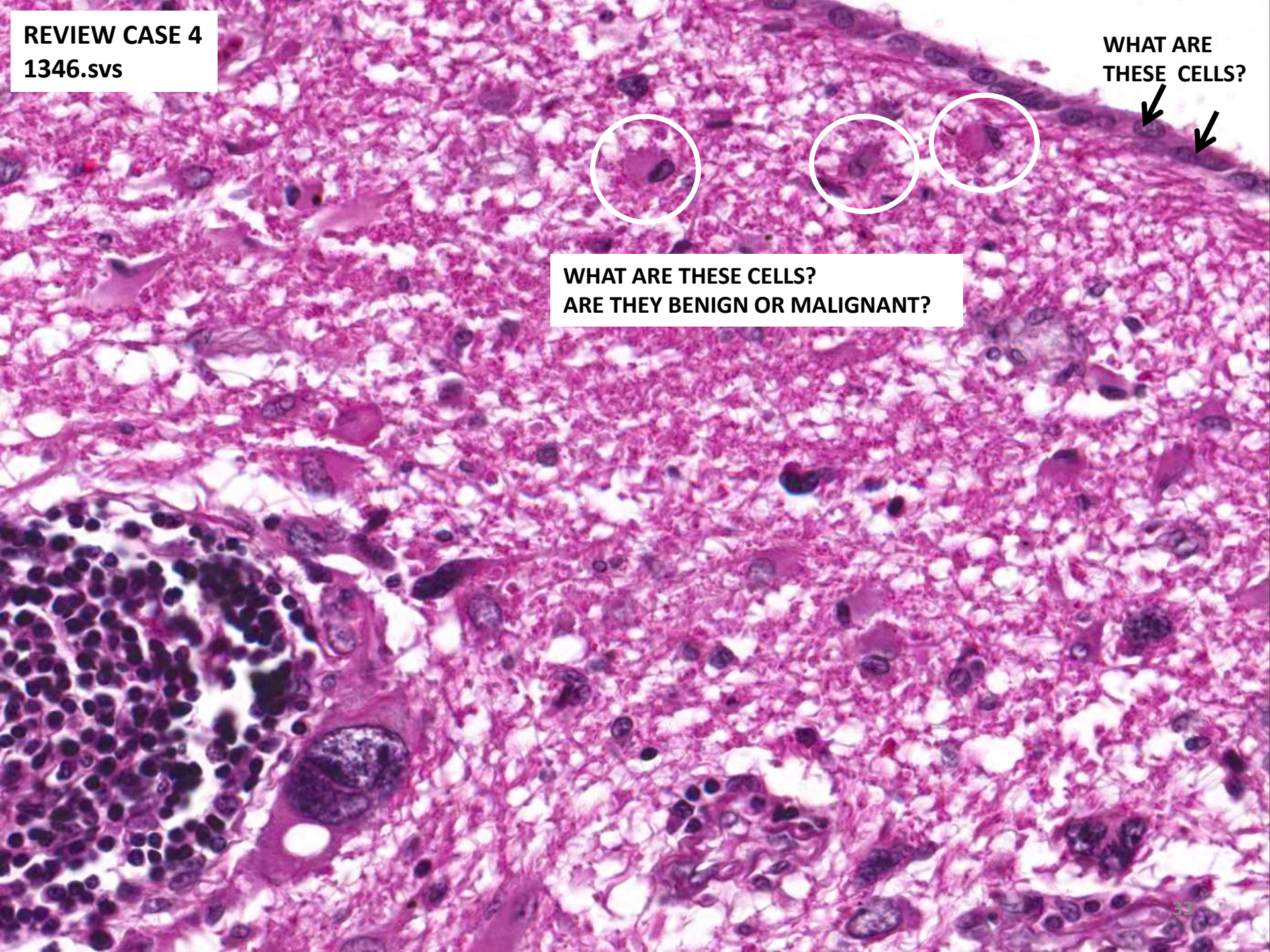


WHAT ARE THE CELLS AROUND THE BLOOD VESSELS?



**WHAT ARE
THESE CELLS?**

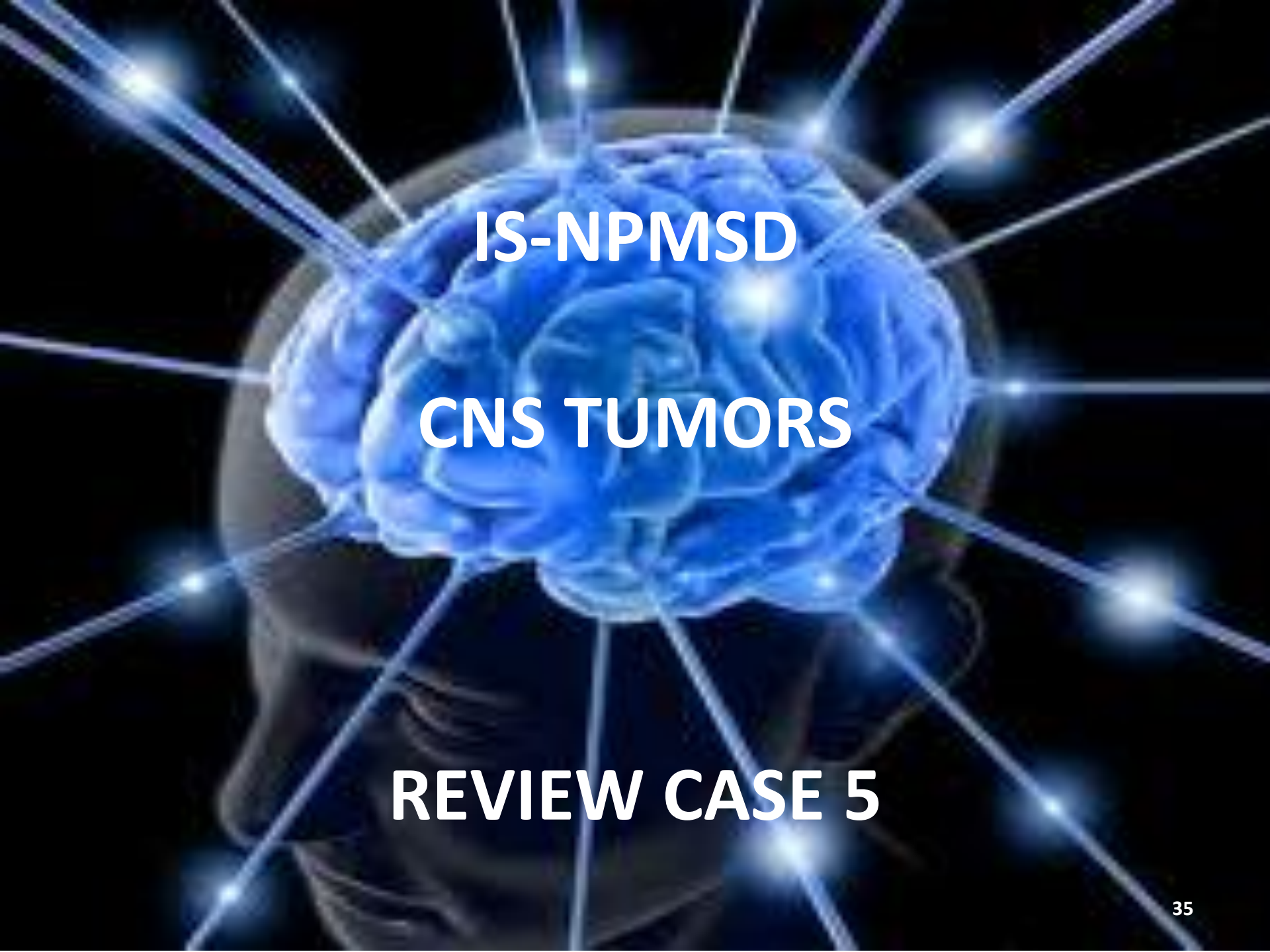
**WHAT ARE THESE CELLS?
ARE THEY BENIGN OR MALIGNANT?**



REVIEW CASE 4
1346.svs

**DESCRIBE THE CHANGE
IN THE NEUROPIL**

**WHAT CELL IS THIS?
IS IT BENIGN OR MALIGNANT?**



IS-NPMSD

CNS TUMORS

REVIEW CASE 5

REVIEW CASE 5

1343.svs

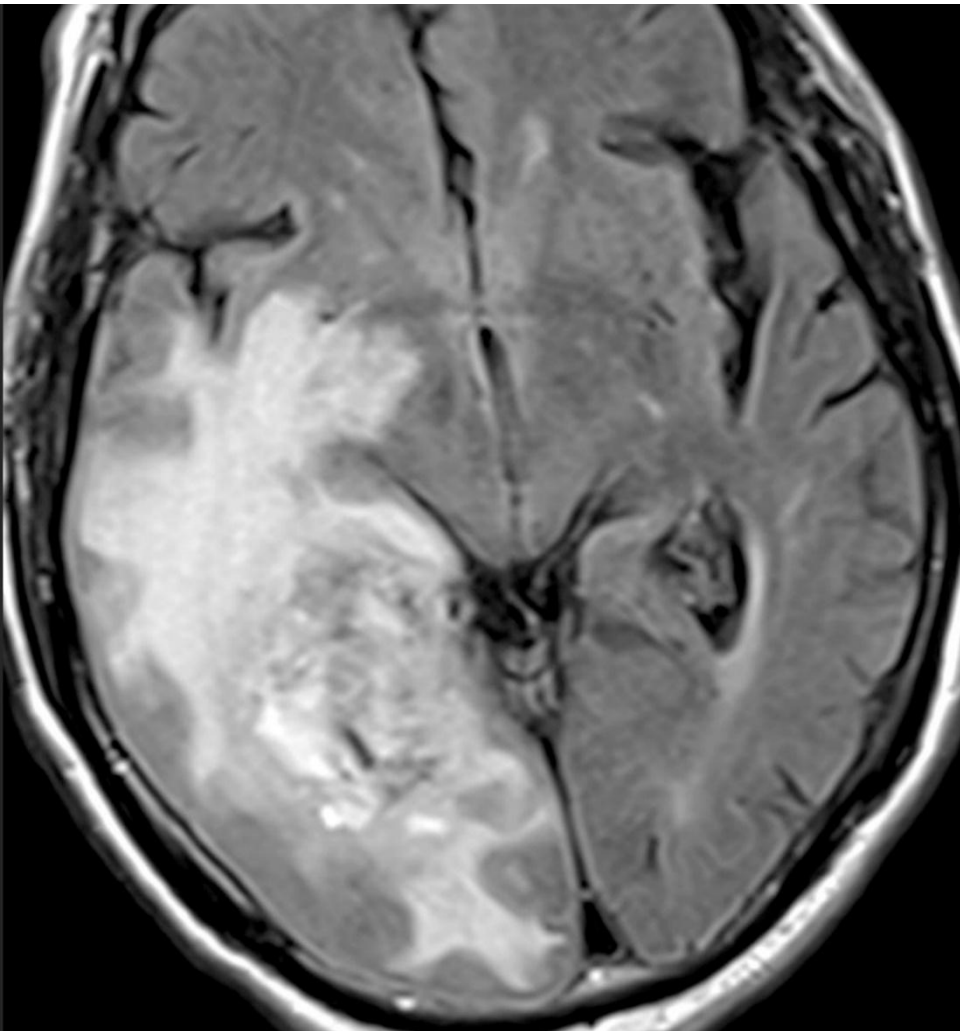
The patient was a 61-year-old man, who developed headache, nausea, and progressive neurologic changes during a 6-month period and was found to have a poorly demarcated mass in the right cerebral hemisphere involving the parietal and temporal lobes. The patient had a partial resection of the tumor followed by radio- and chemotherapy and thalidomide-therapy. After an initial response, the tumor began to grow again, and the patient died 9 months after the initial diagnosis.

1. Is this tumor intra- or extra-dural?
2. intra- or extra-axial?
3. What is your clinical differential diagnosis based on the history and the imaging studies?
4. Is this tumor benign or malignant?
5. What are the diagnostic histologic features?
6. What is the diagnosis?
7. What is the difference in the histologic features between this and the previous case?
8. Why was thalidomide therapy used?

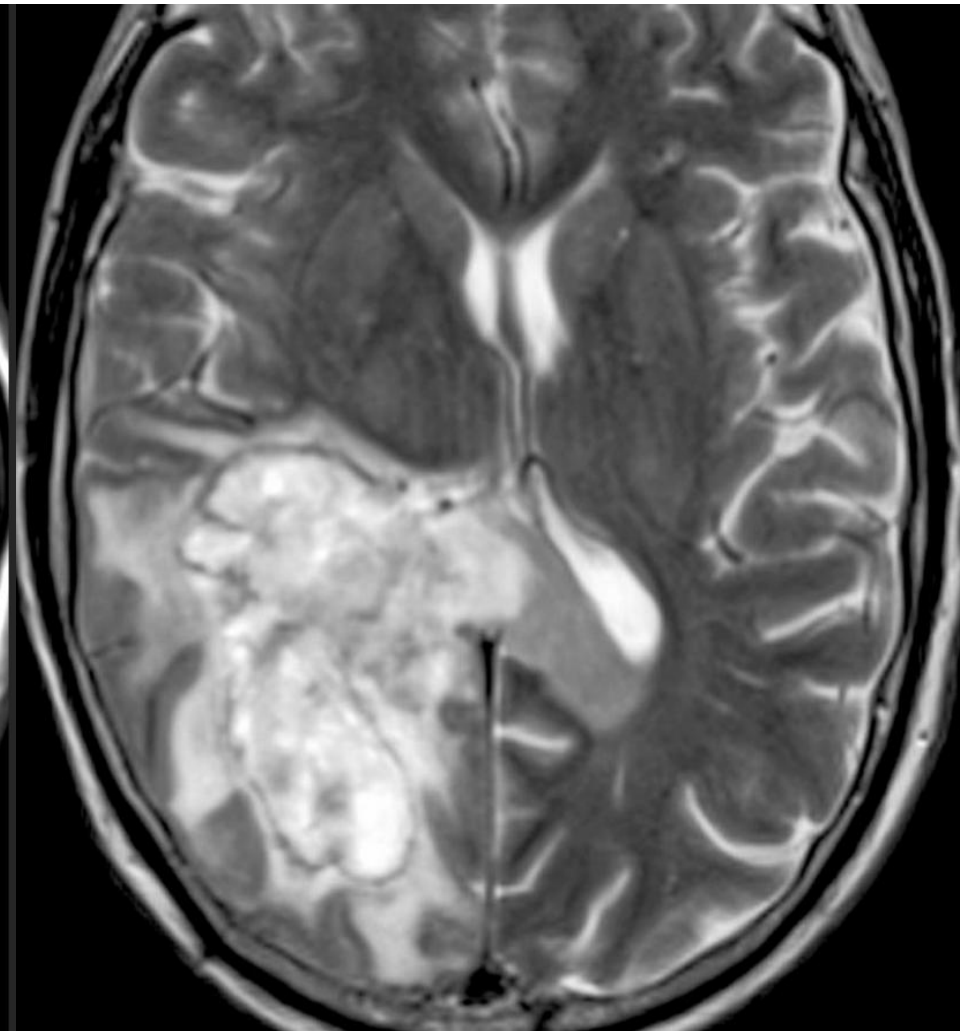
REVIEW CASE 5

MRI WITHOUT CONTRAST

T1



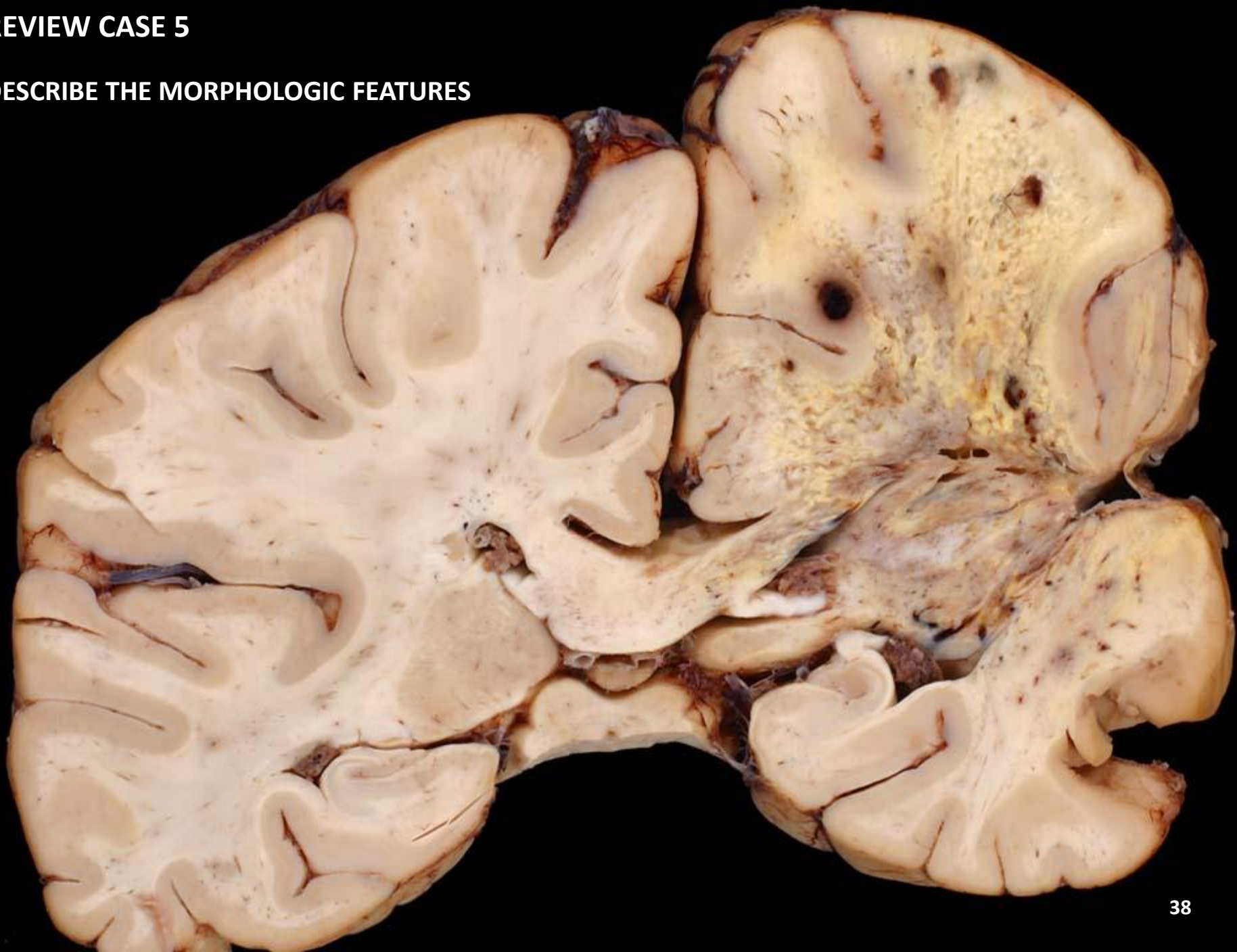
T2



DESCRIBE THE IMAGING FINDINGS.

REVIEW CASE 5

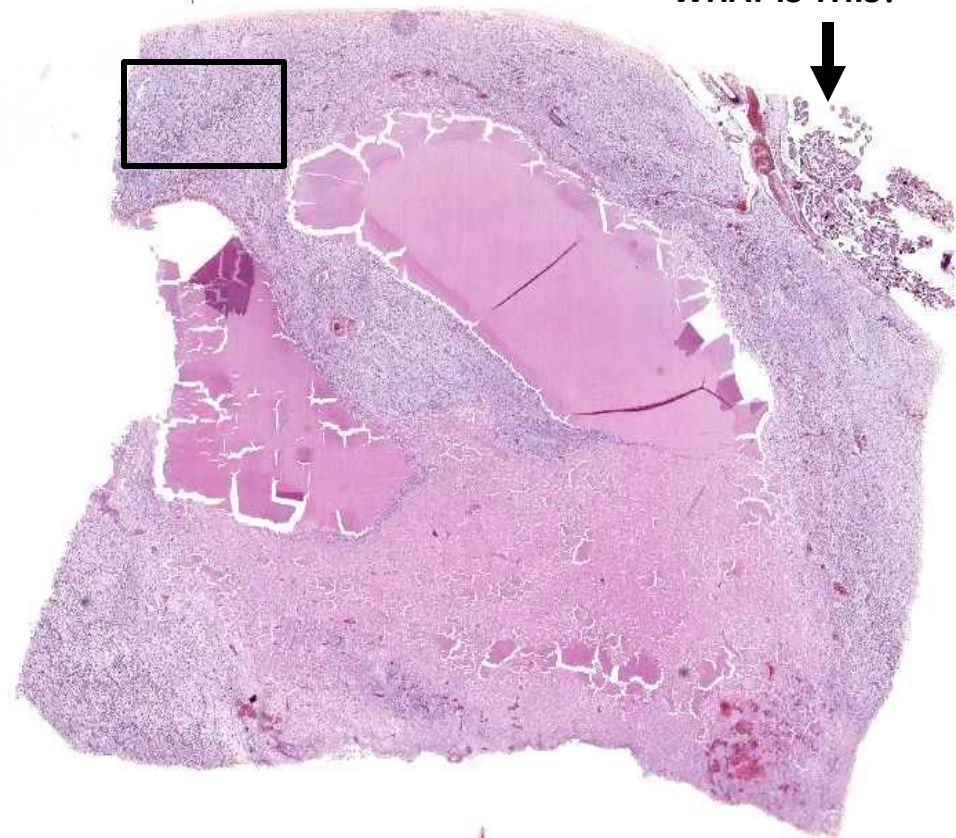
DESCRIBE THE MORPHOLOGIC FEATURES



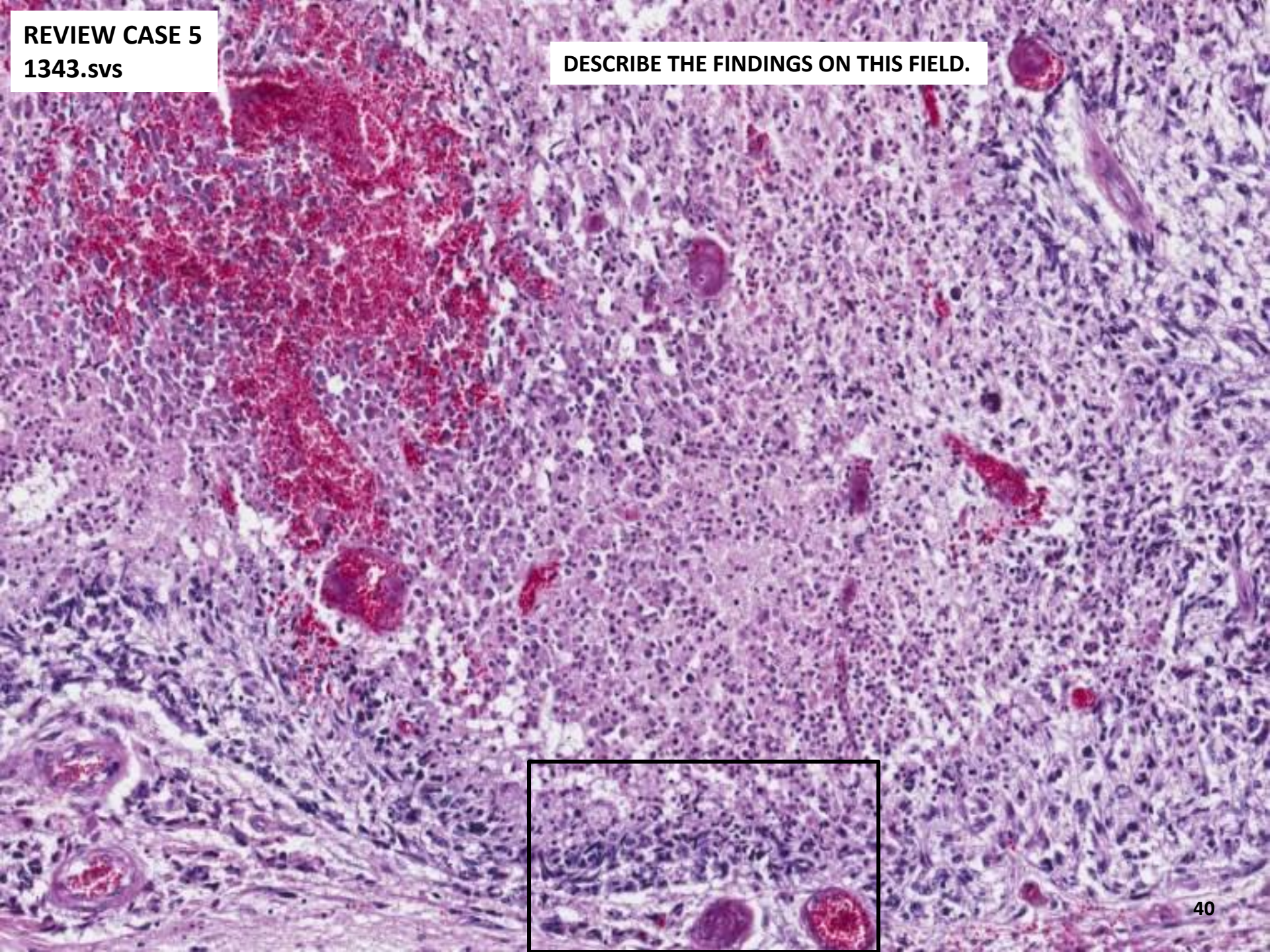
IS THE ARCHITECTURE NORMAL?



WHERE IS THIS?
WHAT IS THIS?



DESCRIBE THE FINDINGS ON THIS FIELD.



REVIEW CASE 5
1343.svs

DESCRIBE THIS AREA

DESCRIBE THIS AREA

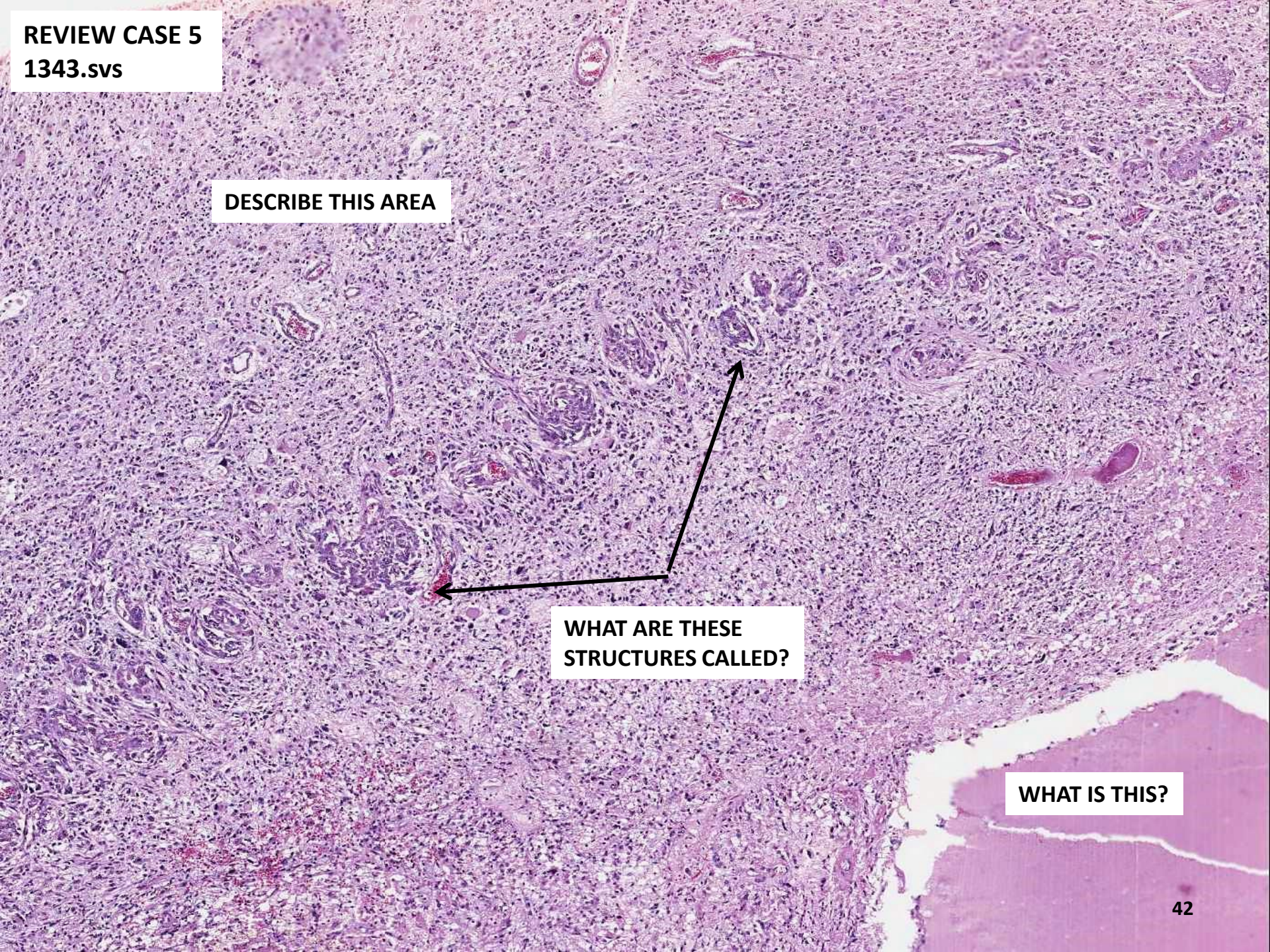
WHAT HAPPENED HERE?

REVIEW CASE 5
1343.svs

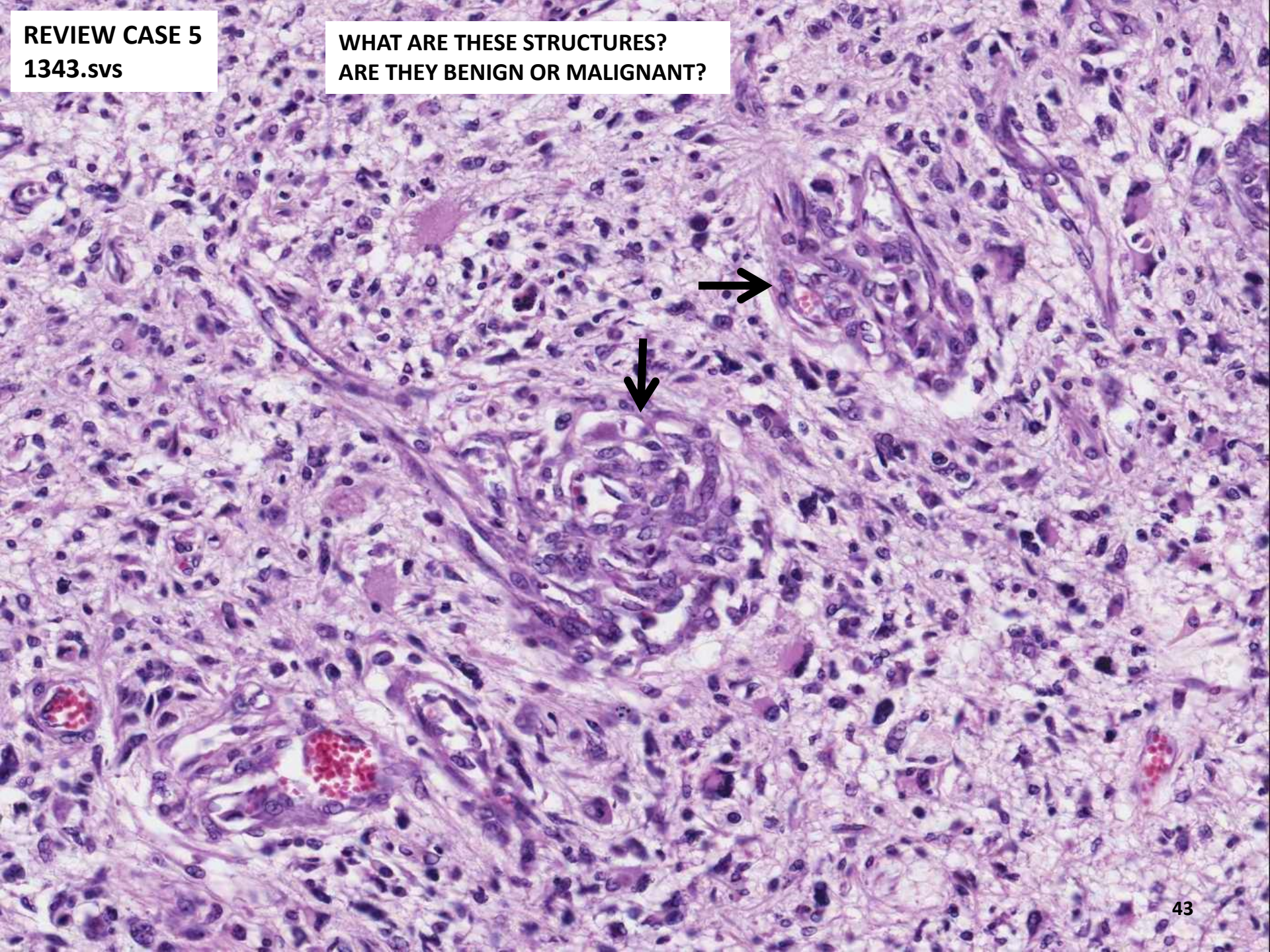
DESCRIBE THIS AREA

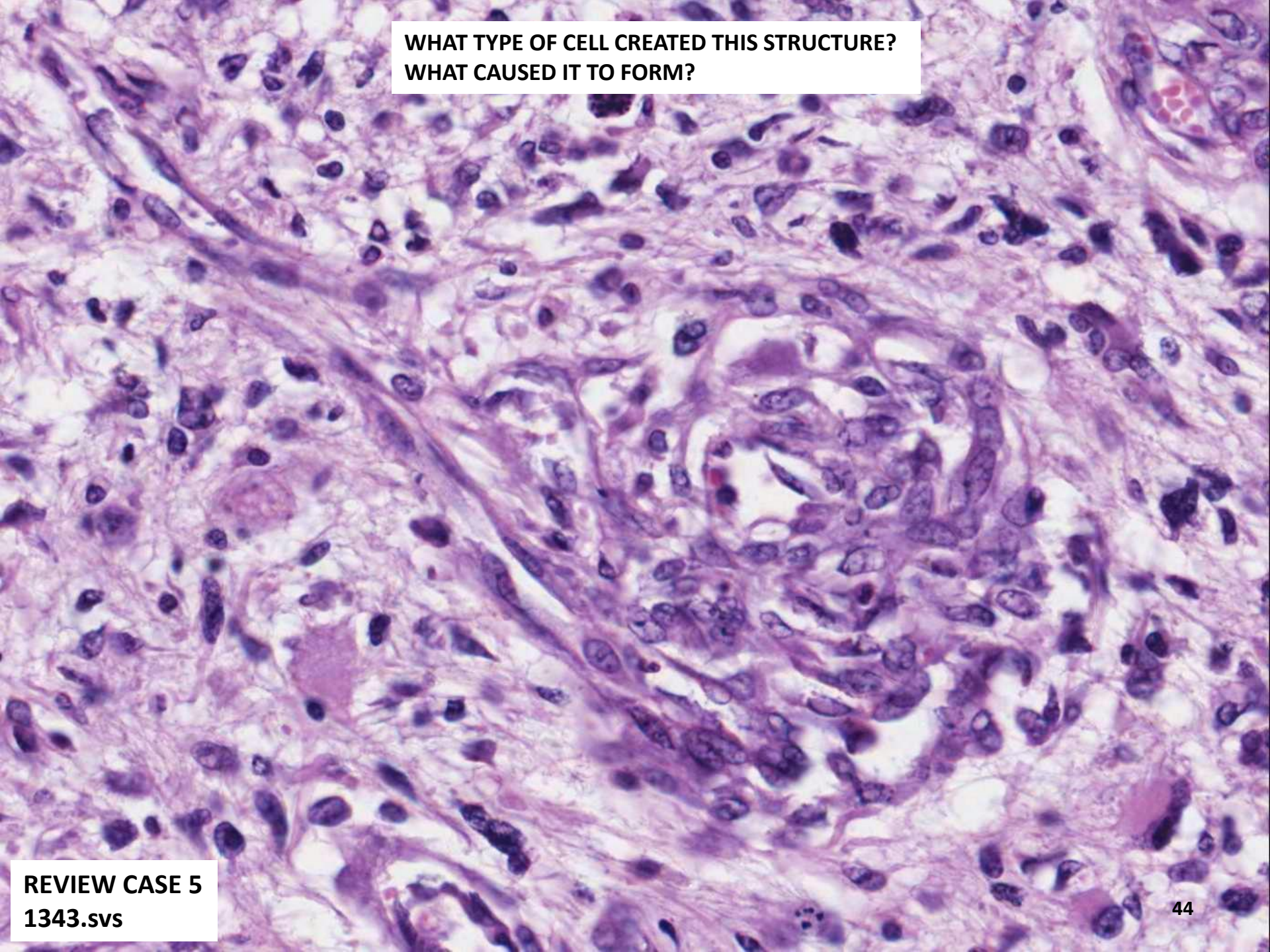
**WHAT ARE THESE
STRUCTURES CALLED?**

WHAT IS THIS?



WHAT ARE THESE STRUCTURES?
ARE THEY BENIGN OR MALIGNANT?





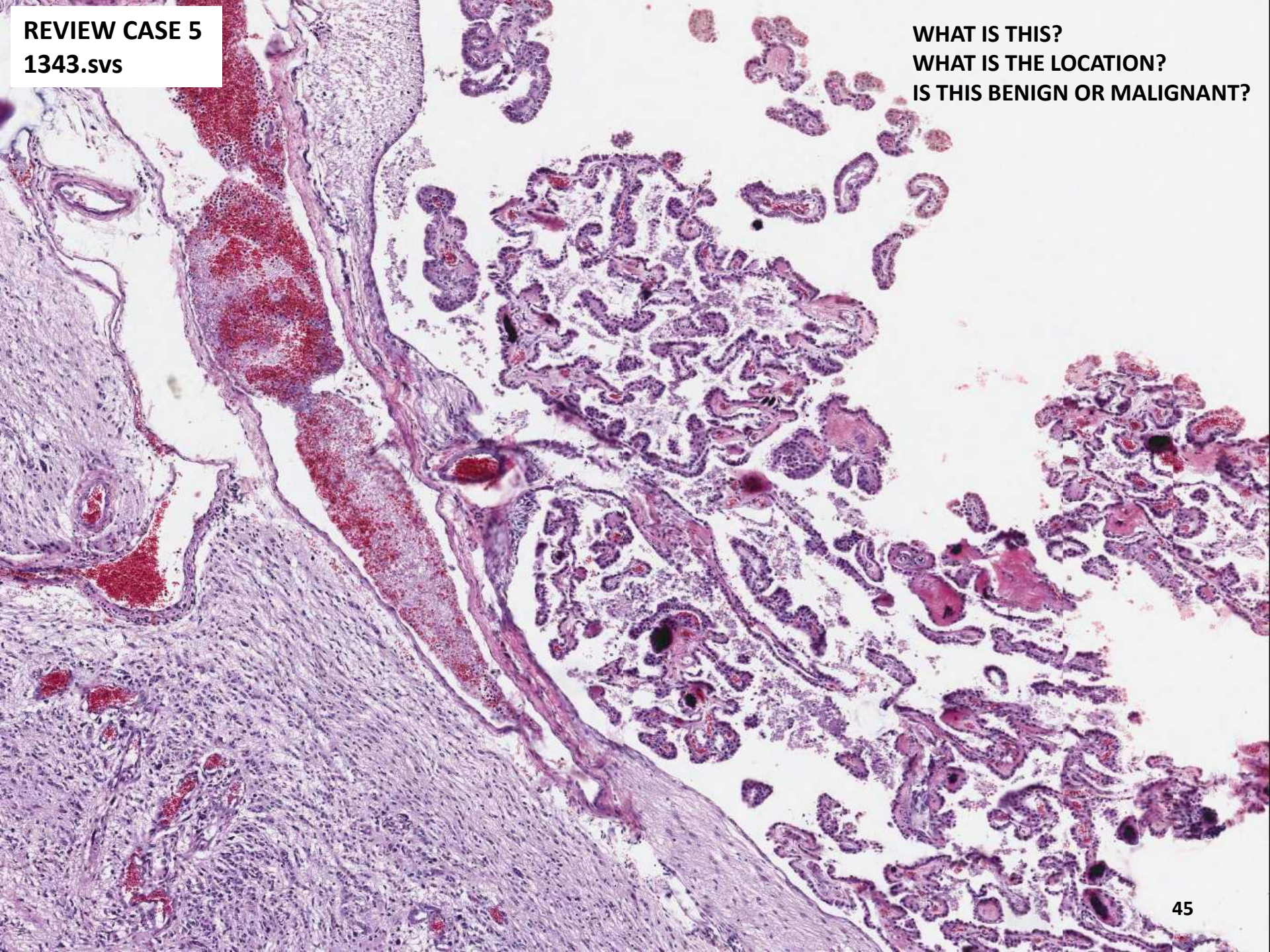
WHAT TYPE OF CELL CREATED THIS STRUCTURE?
WHAT CAUSED IT TO FORM?

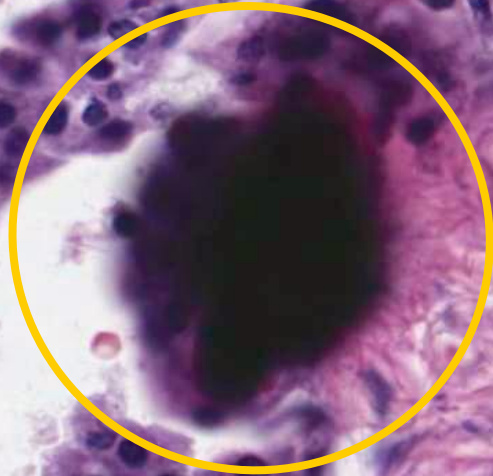
This histological slide shows a tissue section stained with hematoxylin and eosin (H&E). The central feature is a well-defined, circular structure, likely a granuloma or a similar inflammatory response. This structure is composed of a dense collection of cells, including many macrophages (histiocytes) and some multinucleated giant cells. The surrounding tissue is composed of a loose connective tissue stroma with scattered inflammatory cells, including lymphocytes and plasma cells. The overall appearance suggests a chronic inflammatory process, possibly a foreign body reaction or a granulomatous disease.

REVIEW CASE 5

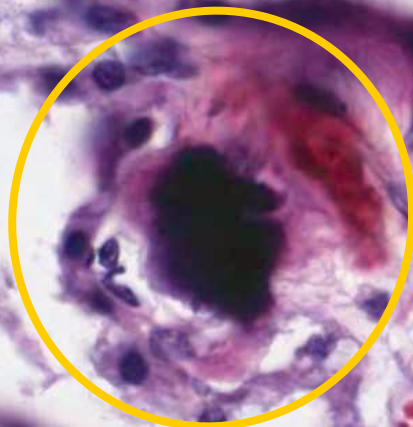
1343.svs

**WHAT IS THIS?
WHAT IS THE LOCATION?
IS THIS BENIGN OR MALIGNANT?**





WHAT ARE THESE?



**ARE THE CELLS
LINING THE
PAPILLARY
PROJECTIONS
BENIGN OR
MALIGNANT?**

IS-NPMSD
CNS AND TUMORS

REVIEW CASE 6

REVIEW CASE 6

1338.svs

A 32-year-old HIV+ female developed lethargy rapid mental deterioration. A CT-scan revealed a supratentorial mass with ring contrast enhancement. A stereotactic biopsy was performed and based on the findings the patient received radio- and chemotherapy. Six months after the initial diagnosis she complained of a severe headache, lapsed into a coma, and died.

1. Is this lesion intra- or extra-dural?
2. Is this lesion intra- or extra-axial?
3. What is the clinical differential diagnosis based on the history and the imaging studies?
4. Is this lesion neoplastic?
5. Is this lesion benign or malignant?
6. What are the diagnostic histologic features?
7. What is your presumptive diagnosis?
8. What studies would you do to make a specific diagnosis?
9. What was the most likely immediate* cause of death?

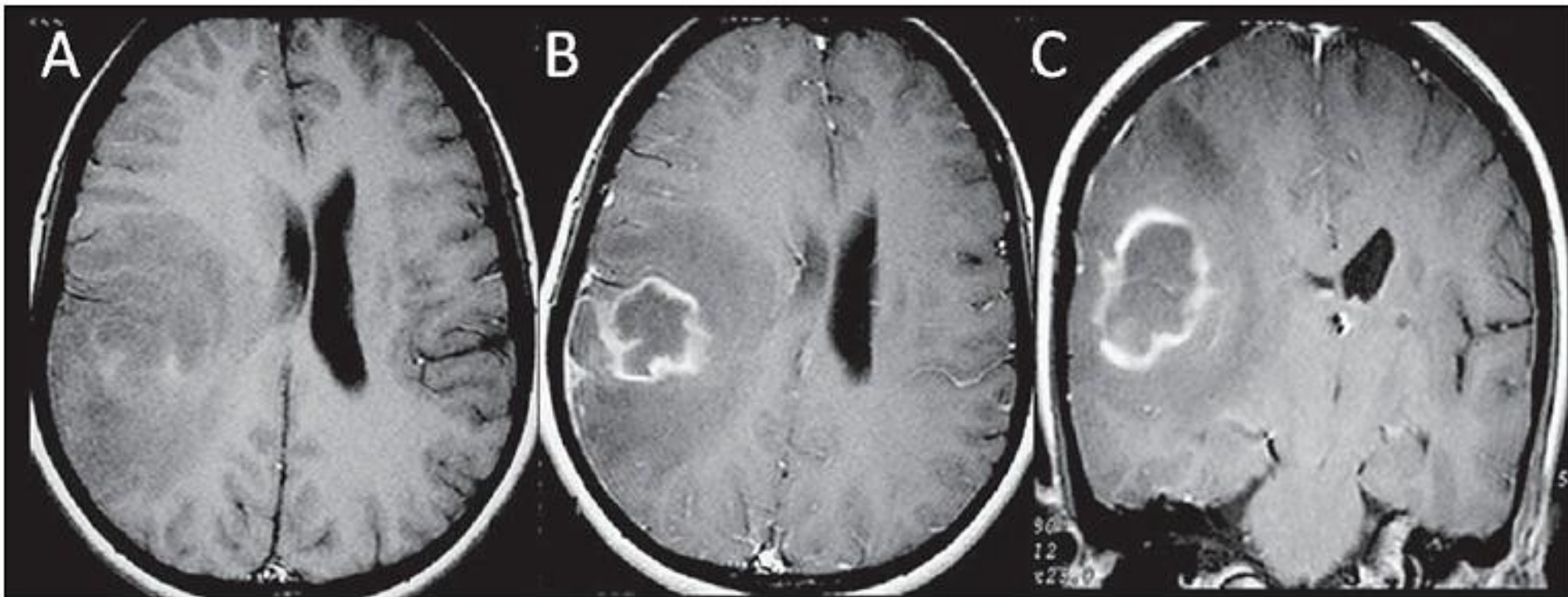
REVIEW CASE 6

MRI

T1-WEIGHTED

T1 WEIGHED POST CONTRAST

T1-WEIGHED POST CONTRAST
CORONAL

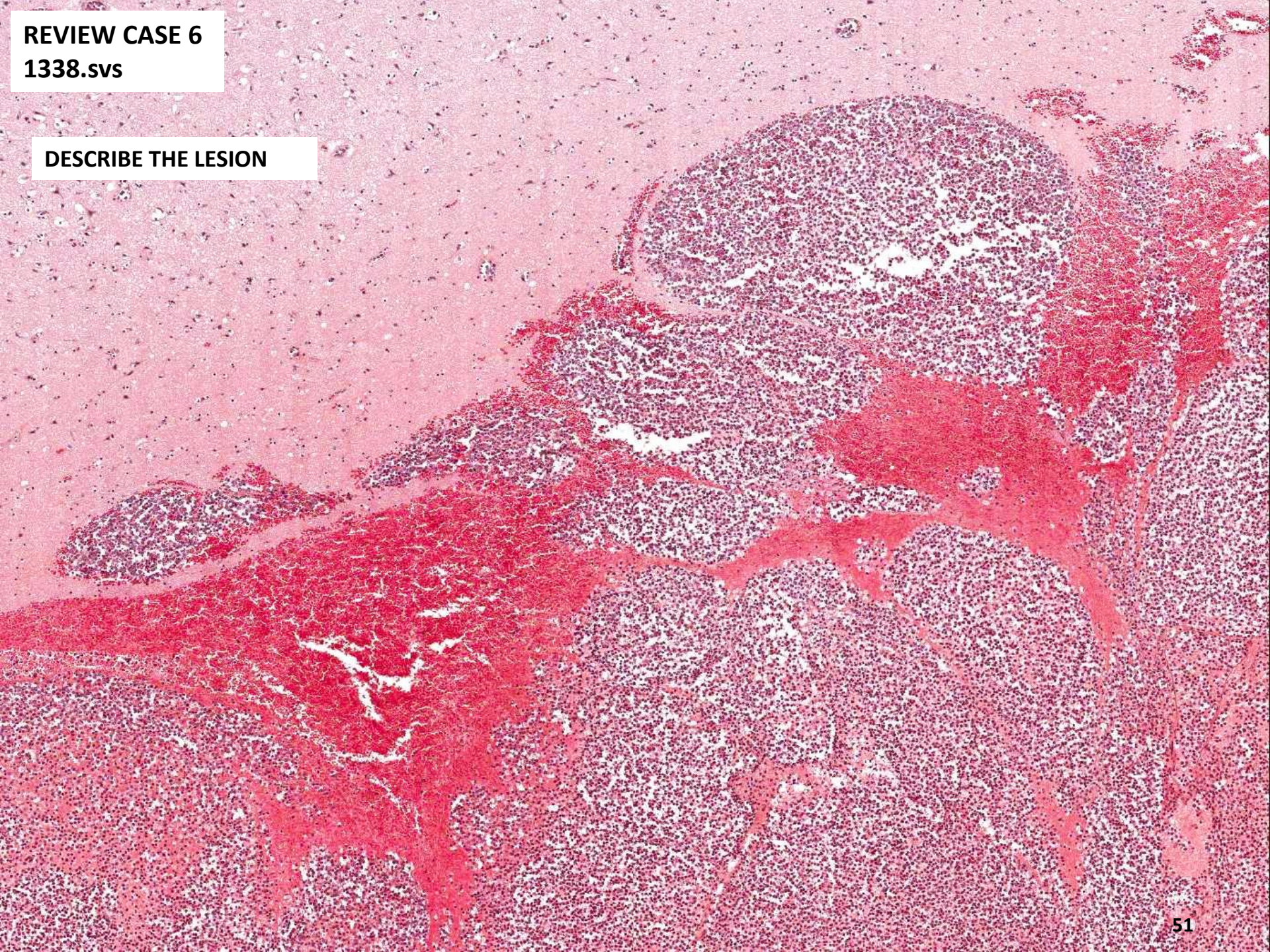


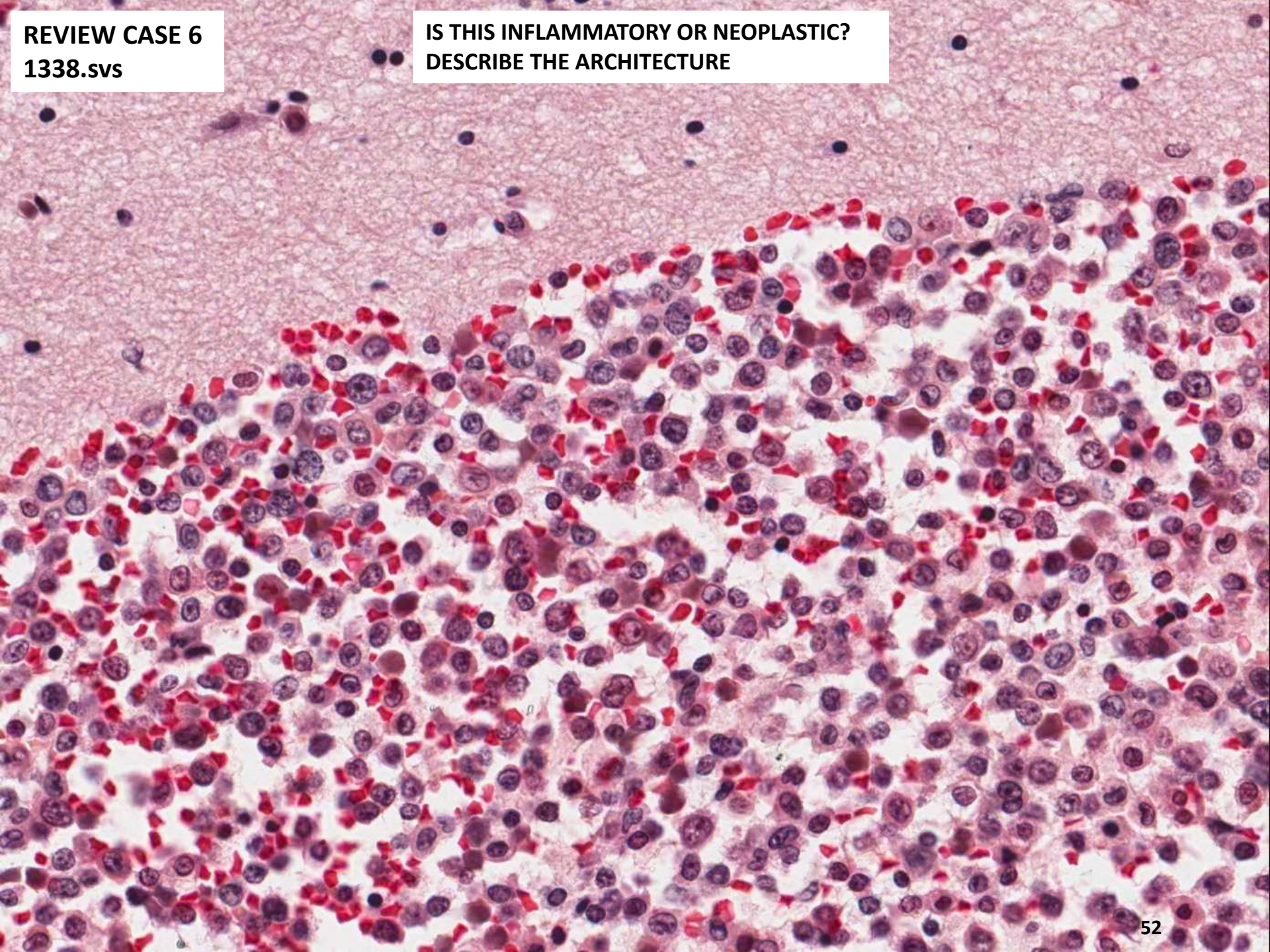
A 32 years old, HIV positive female patient.

DESCRIBE THE IMAGING FINDINGS

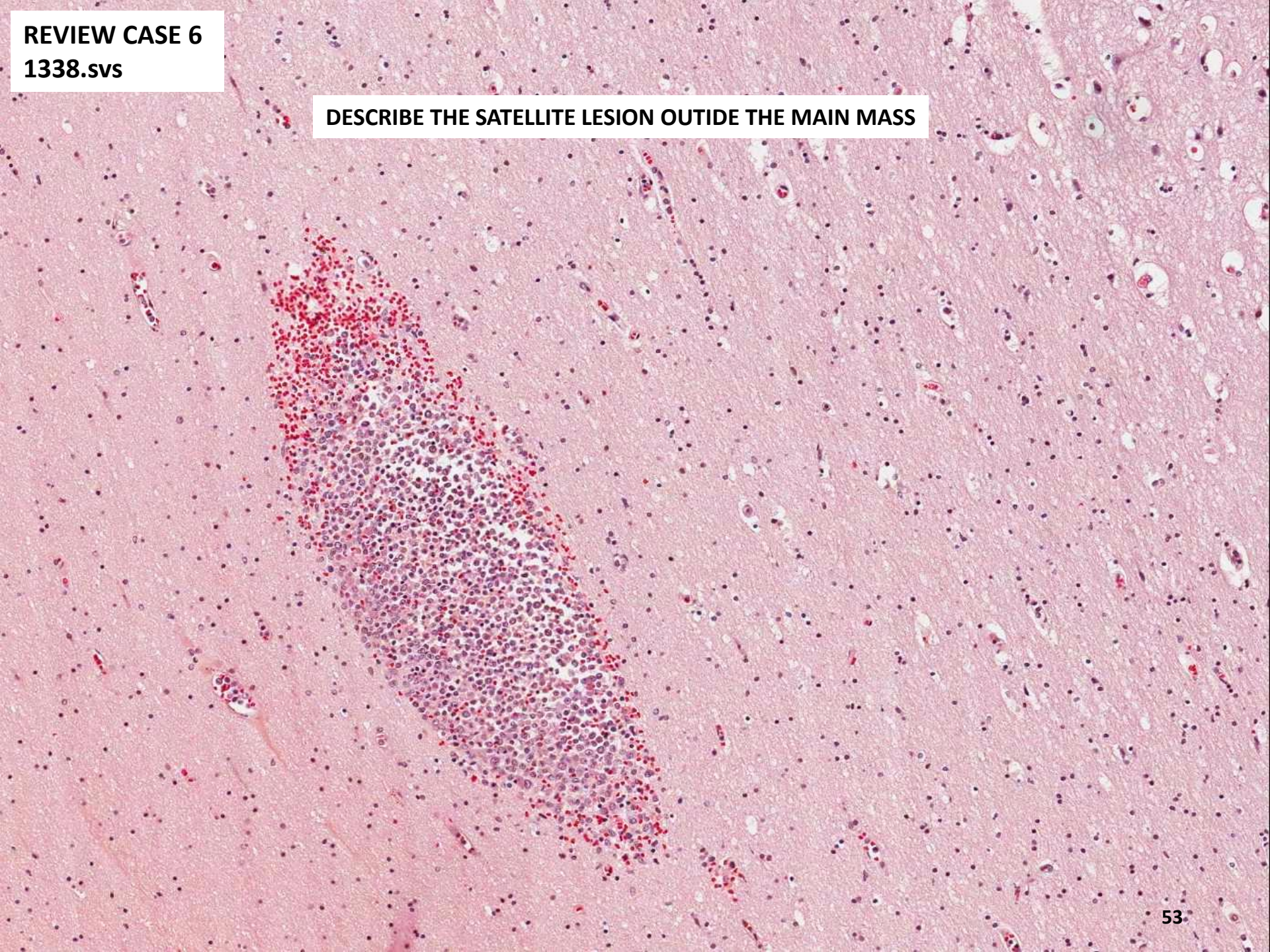


DESCRIBE THE LESION



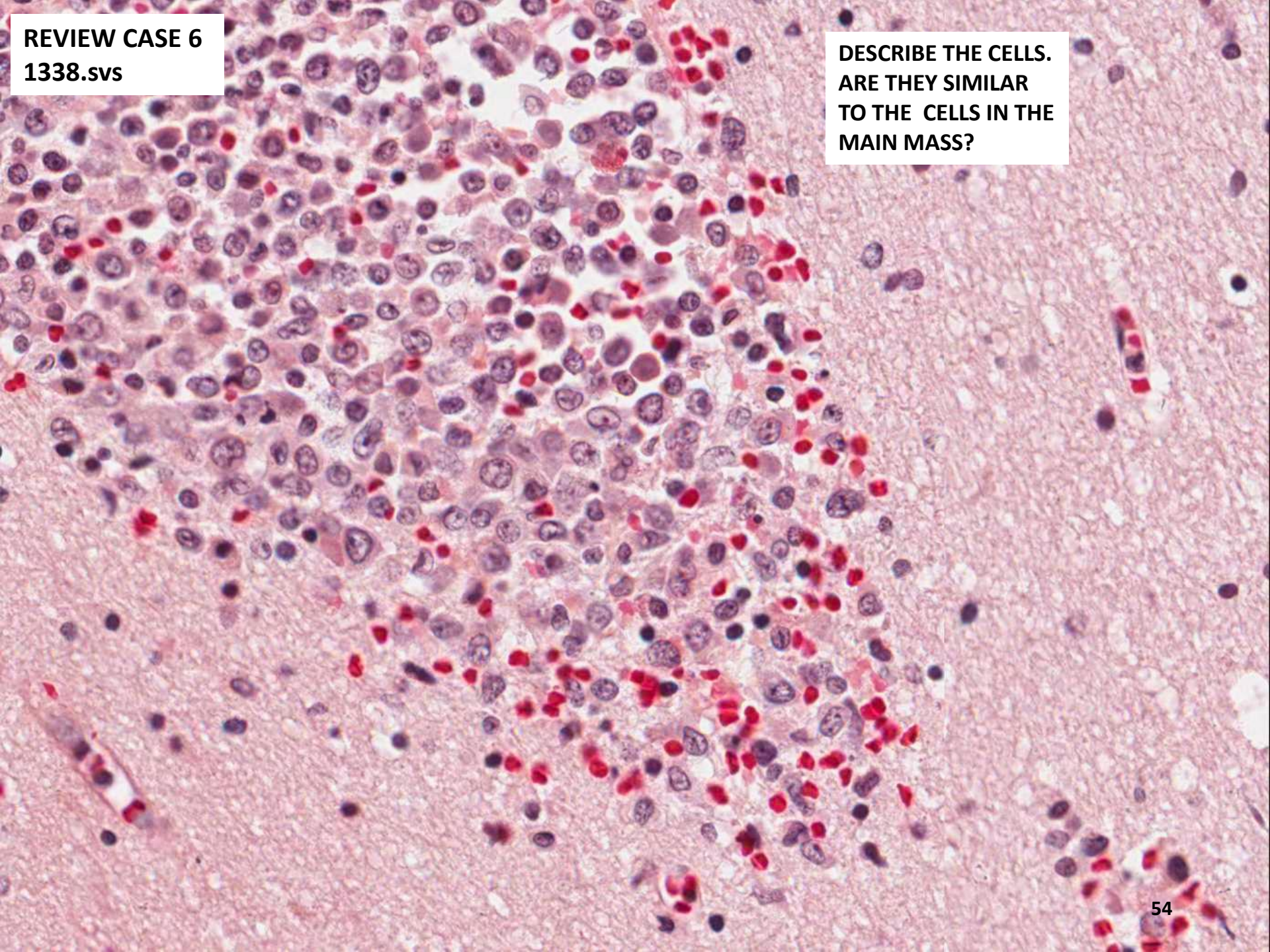


DESCRIBE THE SATELLITE LESION OUTSIDE THE MAIN MASS



REVIEW CASE 6
1338.svs

**DESCRIBE THE CELLS.
ARE THEY SIMILAR
TO THE CELLS IN THE
MAIN MASS?**





QUESTIONS?